

TRANSACTION SECURITY



Transaction Security in BASE24-eps

- BASE24-eps supports the following transaction security functions:
 - PIN encryption
 - PIN verification
 - Message authentication (MACing)
 - Card verification
 - Dynamic Card Verification
 - Chip authentication
 - Automated Key Distribution System
 - EMV processing
 - Contactless Magnetic Stripe
 - Dynamic Key Change support



Transaction Security in BASE24-eps

- BASE24-eps supports the following transaction security functions:
 - Two or Three Level Key Hierarchy
 - Previous, Current and Future Key support
 - Series key support for Interfaces
 - Data Encryption for account balances (Interac)
 - German Market Security support (ZKA Specifications)
 - French Market Security support (NAC Specifications)
 - UCAF Validation (MasterCard Secure Code transactions, limiting card not present risks associated with internet purchases)



Cryptographic Keys

- Customers may need to enter a PIN when initiating an EFT transaction. This PIN gets encrypted at the device and remains encrypted until it is verified by the issuer.
- Keys used to encrypt PINs are called PIN Encryption Keys; session key, B-Key, transaction key, temporary key, working key, or communication key.
- These keys can be exchanged manually or thru an electronic message exchange.
- If exchanged electronically, the key itself will be encrypted before it is exchanged.



Key Hierarchy

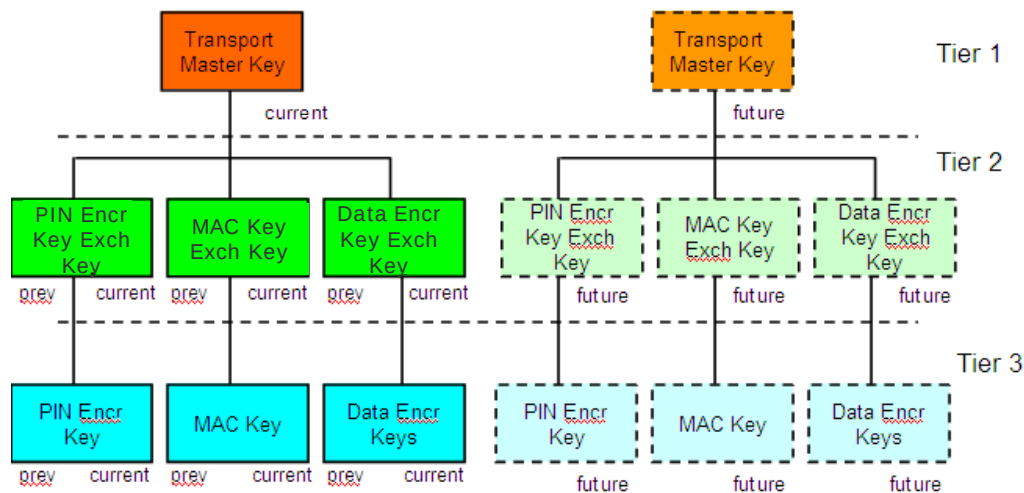
- 2-Level Key Hierarchy (default)
 - Supports Key Exchange Keys (KEKs) manually exchanged for Interfaces, for ATMs, can support AKDS to dynamically exchange the ATM KEK (Terminal Master Key)
 - Working Keys – dynamically exchanged, PIN encryption key, MAC key, Data encryption key
- 3-Level Key Hierarchy*
 - Supports a higher level key – Transport Master Key (TMK)
 - Exchanged manually or via AKDS if supported
 - Key Exchange Keys (KEKs) – dynamically exchanged
 - Supports a single KEK for all working keys or separate KEKs for each key type (PIN, MAC, DATA)
 - Working Keys – dynamically exchanged
 - PIN working key, MAC working key, Data working key

*Atalla AKB, Eracom Mark II, IBM Crypto Card, Bull BNTng



Future Keys (KEKs)

- Manually entered in advance of use, but not active
- A key verification message is generated by an operator to verify both parties have successfully loaded the correct (same) key
- Key is activated at an agreed time
- New Data source CSECF/ISECF for future keys
- Required by Faster Payments Service (FPS)
- Supported by Atalla AKB, Bull BNTng, IBM Crypto Express 3, SafeNet (Eracom) ProtectHost White Mark II, and Thales e-Security (Racal) HSMs



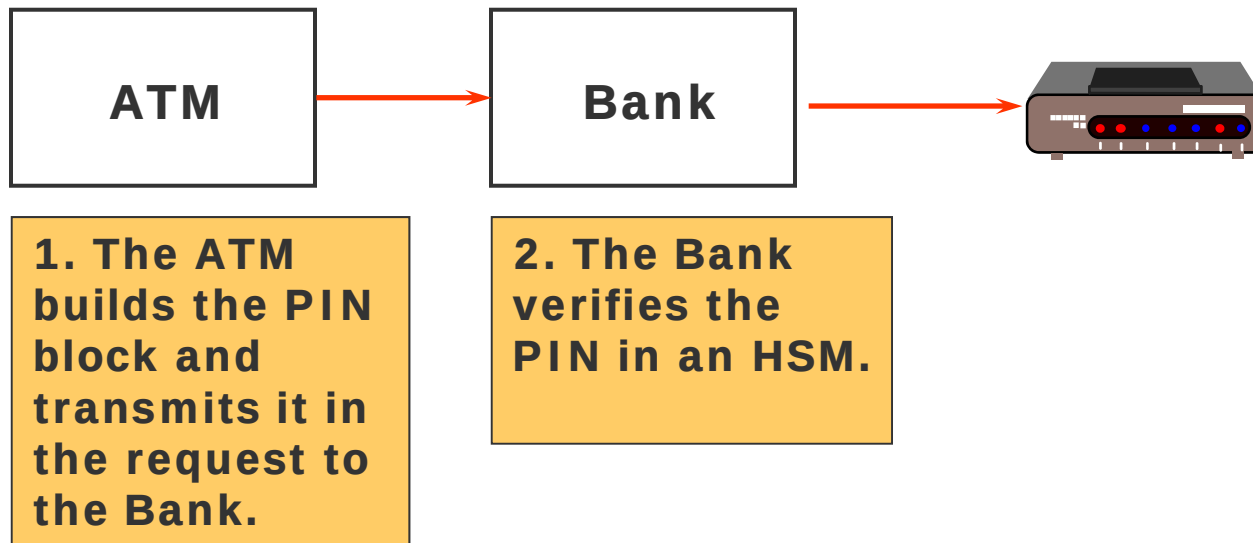
Hardware Security Modules (HSM)

- All cryptographic functions occur within a secure hardware security module (HSM).
- HSM vendors include:
 - Atalla
 - Thales
 - SafeNet (Eracom)
 - Bull
 - IBM Crypto Express
- These HSMs are initially set up with their own encryption key which will be used to encrypt every key stored in the BASE24-eps database, HSM vendor database, or HSM itself.



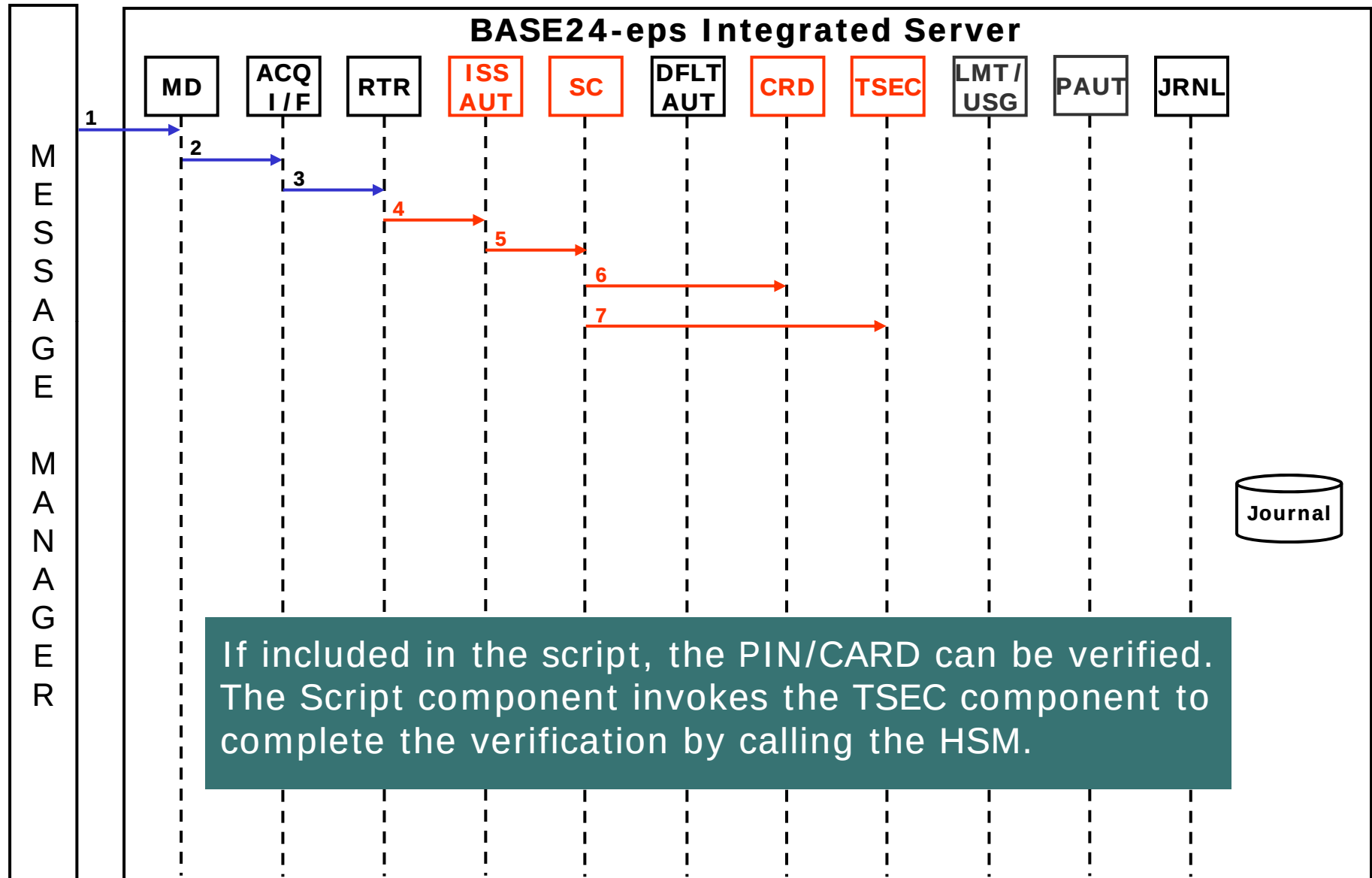
Processing Encrypted Data

- Example: PIN Block



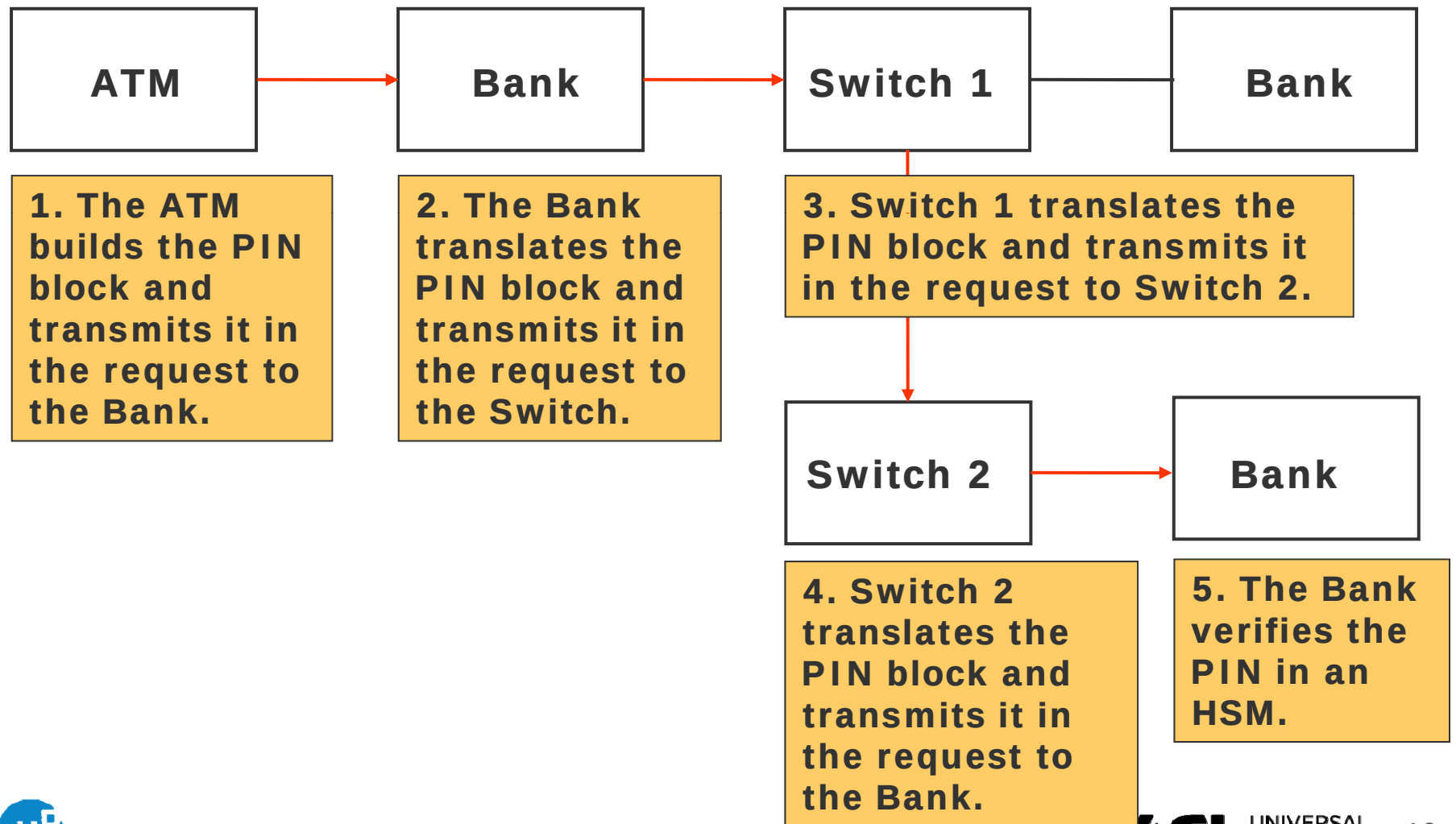
- No sensitive data is returned to the device.

Transaction Processing

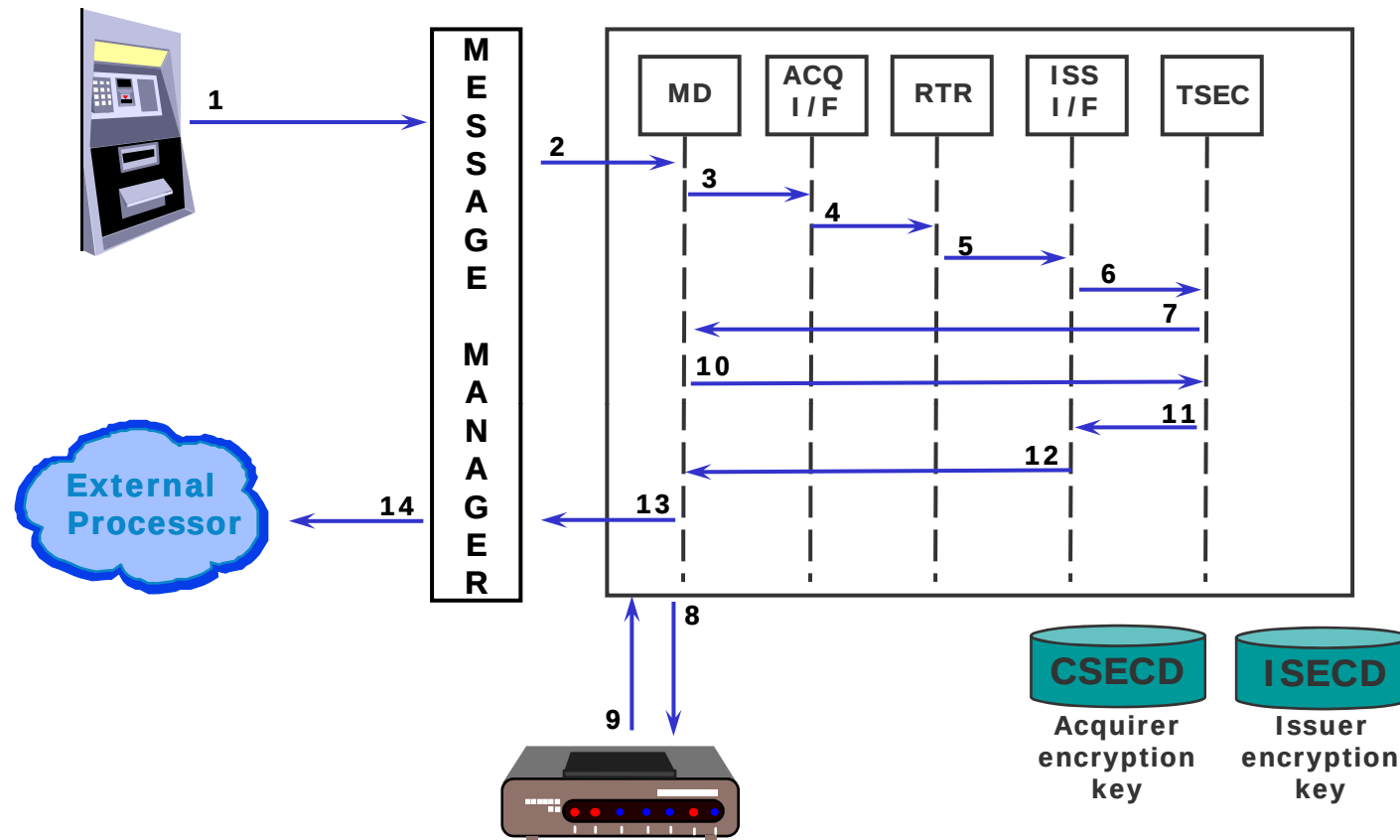


Processing Encrypted Data

- Example: PIN Block

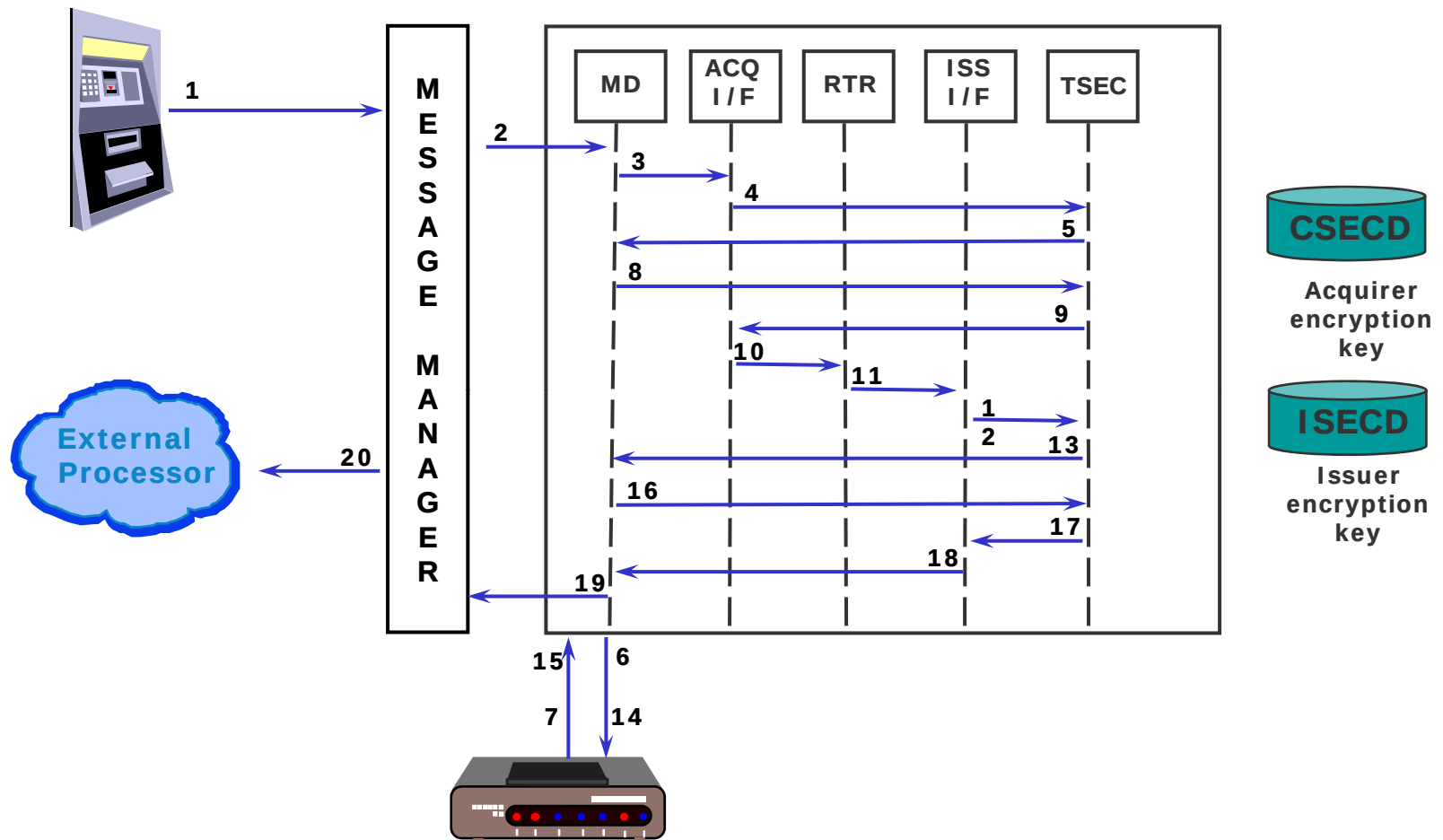


Translation



The Issuer Interface component is responsible for making the call to translate the encrypted PIN block from the acquirer key to the issuer key before sending the request.

Message Authentication



The Acquiring Interface will process the MACing requirements on the request. If the request needs to be sent to an external processor for authorization, the Issuer Interface may also need to process MACing on the request. This will also occur when processing the response.

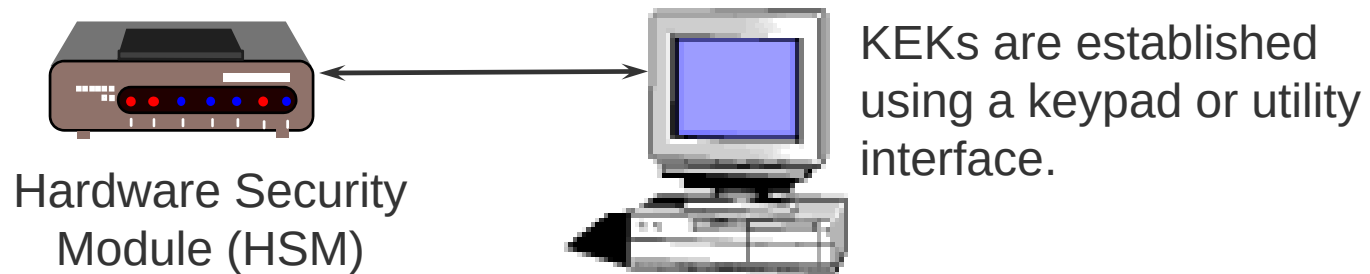


TRANSACTION SECURITY

Basic Configuration Thales/Atalla

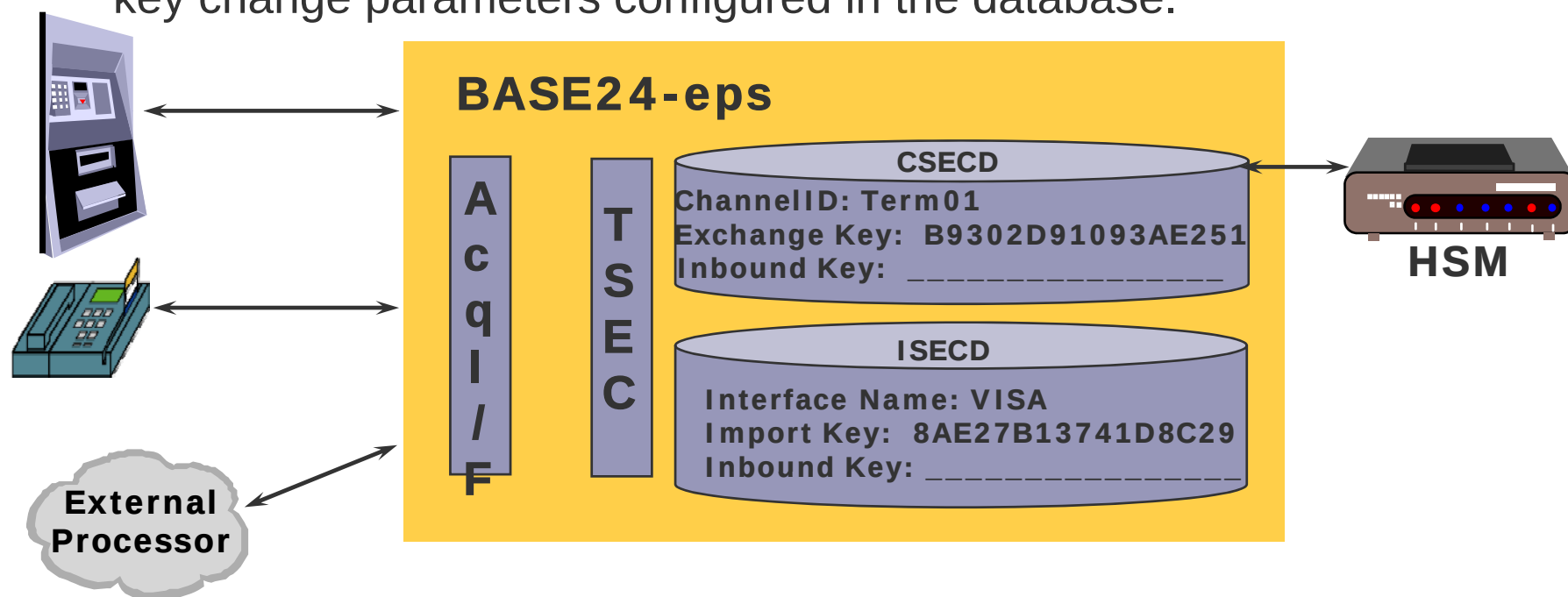
Terminal Master Keys/KEKs

- ATM/POS devices and external processors will use an exchange key for programmatic delivery of PIN/MAC encryption keys.
- Generally, initial setup of the exchange key is a manual process. ATMs do support a programmatic delivery of this exchange key (Master Key or A-Key).



Dynamic Key Change

- Once the exchange keys are established (with the external processor (KEKs) or the device (Terminal Master Keys)), then PIN/MAC keys can be exchanged dynamically based on dynamic key change parameters configured in the database.



If the external processor does not support a dynamic exchange of the encryption keys, then the KEK is not used.



TRANSACTION SECURITY

Basic Configuration IBM System Z



Trusted Key Entry (TKE)

- In a live environment, a Trusted Key Entry (TKE) workstation is used to generate keys.
- The TKE workstation supports chip cards providing the interface required to create and manage new keys.
- The TKE workstation interfaces with the Integrated Cryptographic Service Facility (ICSF) subsystem.



Key Labels

- The key label is stored in the CSEC/ISEC data source.
- Key labels will be derived based on the data source(s) primary key coupled with a system defined key type*.

– Example:

- Channel Inbound PIN key

- Channel data source primary key: ATMTERM01
- System defined key type: TPK_IN

TPK.IN.ATMTERM01

* The system defined key types are listed in the Transaction Security manual



Key Label Examples

- Channel Security data source (CSECS)

Key Description	Key Tag	Primary Key	Key Label
PIN Master Key	TMK_PIN	Channel ID	TMK.PIN.Channel ID
Inbound PIN Key	TPK_IN	Channel ID	TPK.IN.Channel ID

- Interface Security data source (ISECD)

Key Description	Key Tag	Primary Key	Key Label
Inbound PIN Exchange Key	ZMK_PIN_IN	Interface Name	ZMK.PIN.IN.InterfaceName
Outbound PIN Encryption Key 1	KPE_OUT1	Interface Name	KPE.OUT1.InterfaceName

All Key Labels are defined in the Transaction Security Manual



Terminal Master Key

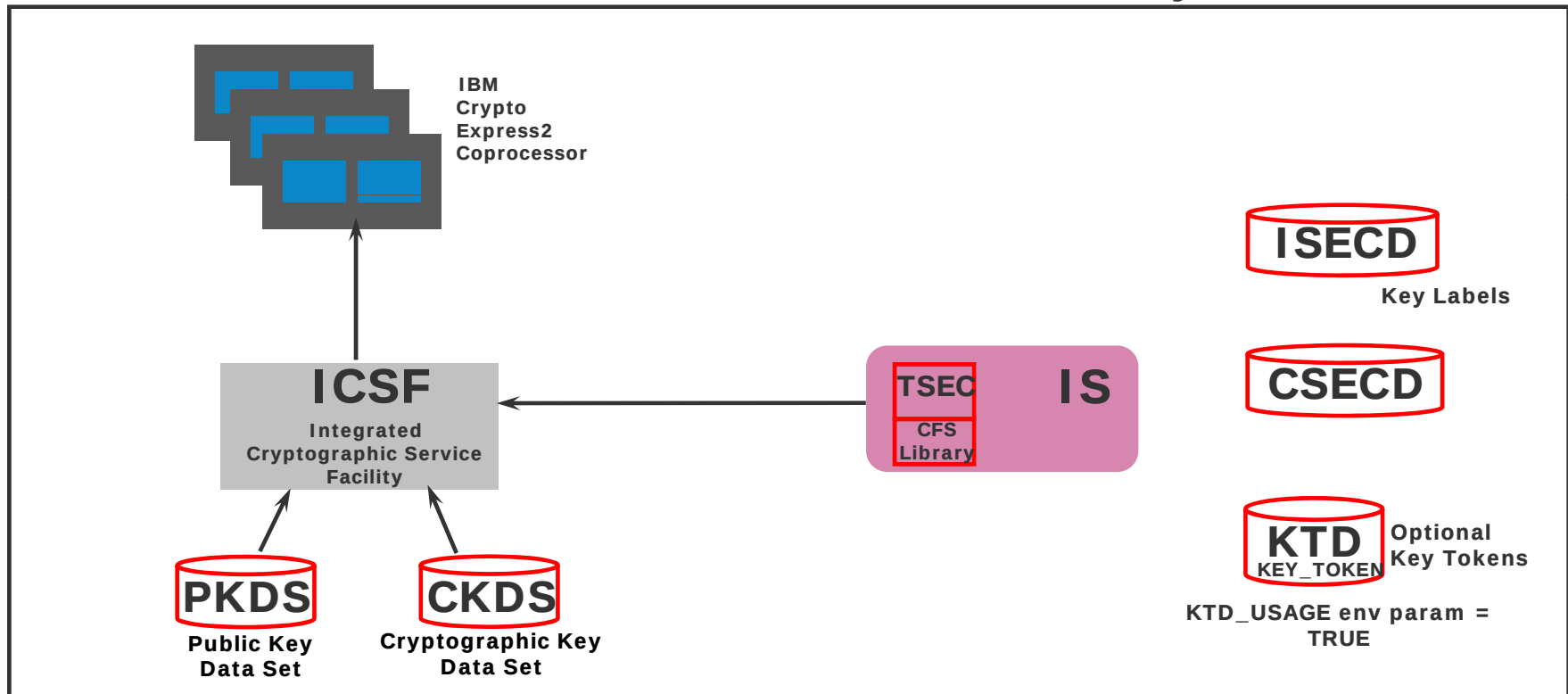
- Using ICSF, generate 2 random components in the clear.
- Load the components into the CEX2 registry (CEX2R) chip card using TKE workstation along with the key label.
- Initiate the load operational key function using the ICSF TKE panel which moves the components from CEX2R to CKDS.



Terminal Master Keys/KEKs

- The IBM Crypto Express2 chip will process all cryptographic functions

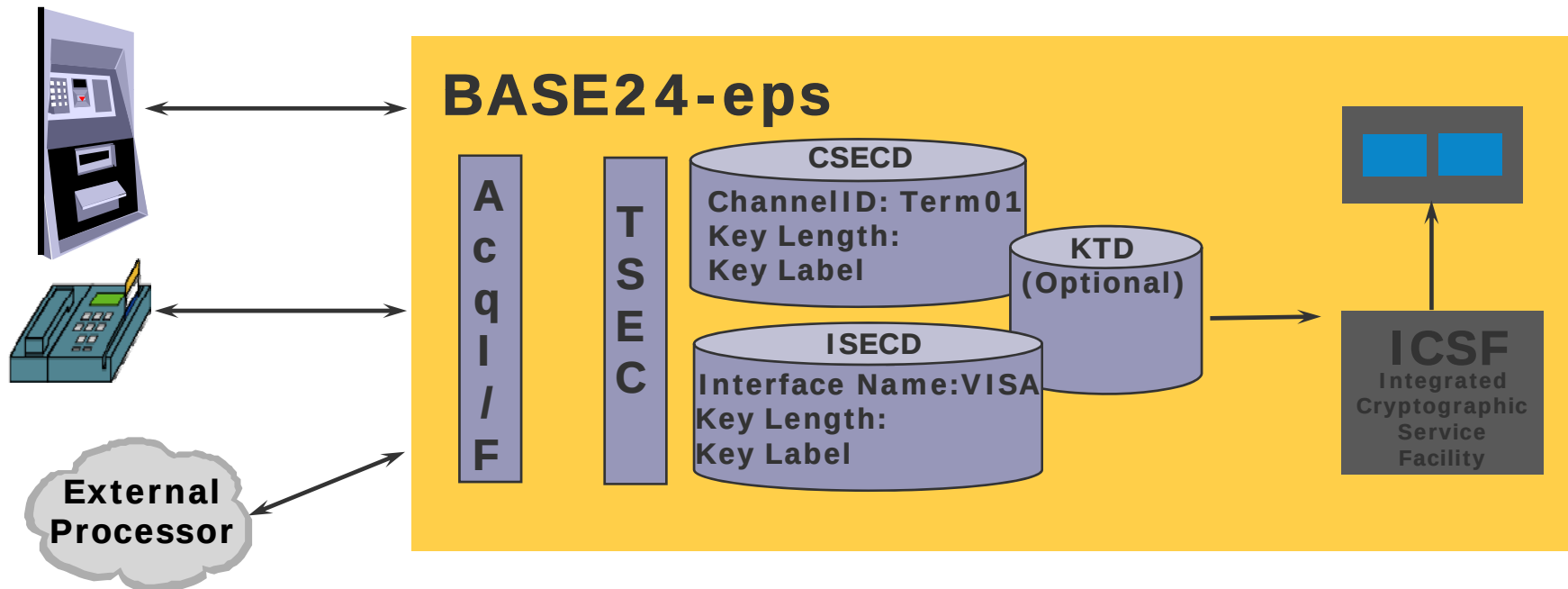
IBM System z



- The KTD key token data source contains public keys if AKDS is supported. For device populations > 70,000, it is recommended the KTD be used to store the dynamic keys.

Dynamic Key Change

- Once the exchange keys are established (with the external processor (KEKs) or the device (Terminal Master Keys)), then PIN/MAC keys can be exchanged dynamically based on dynamic key change parameters configured in the database.

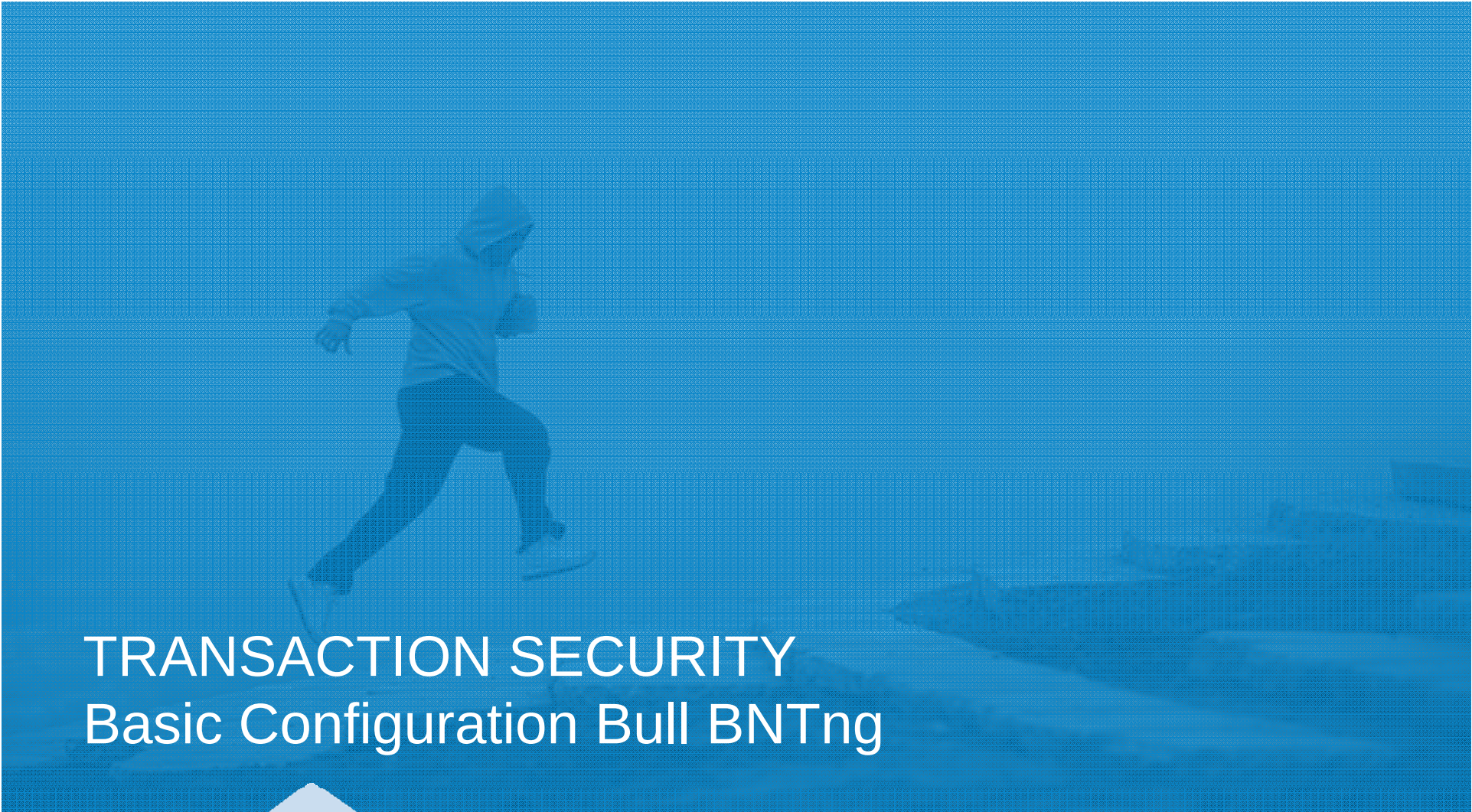


If the external processor does not support a dynamic exchange of the encryption keys, then the KEK is not used.

3 Level Hierarchy

- IBM Crypto Chip also supports the 3 level hierarchy key scheme
- Level 2 keys are stored in KTD (Key Token data source)
- Device Handler components and Interface components require additional development





TRANSACTION SECURITY

Basic Configuration Bull BNTng

Transaction Security Configuration

- ATM/POS devices and external processors will use an exchange key message for programmatic delivery of PIN/MAC encryption keys.
- Initial setup of keys is performed using a Key Loader process. Multiple CGDC binary input files, provided by Bull Key Management Centre (KMC), are used to load static keys for storage on the BASE24-eps database.
- Key header creation
- Key types include:
 - general keys (KDK) for key distribution and storage for a unique HSM
 - acquirer keys (e.g., RSA, KME, KTK, KT,...)
 - issuer keys (e.g., IMKcvc3, KCVX, KPV, ...)
 - EMV keys (e.g., IMKac, IMKsm, ...)

Transaction Security Configuration

- Support for the French marketplace transaction security requirements using Bull BNTng HSM. Functionality includes:
 - key exchanges (key encrypting keys, PIN encrypting keys, MAC encrypting keys)
 - configuration and calculation of MAA authorization number
 - up to 10 VISA PVV Keys per Issuer BIN and Expiry Date
 - MasterCard AAV Validation (SPA/UCAF)
 - VISA DCVV and MasterCard CVC3 validation



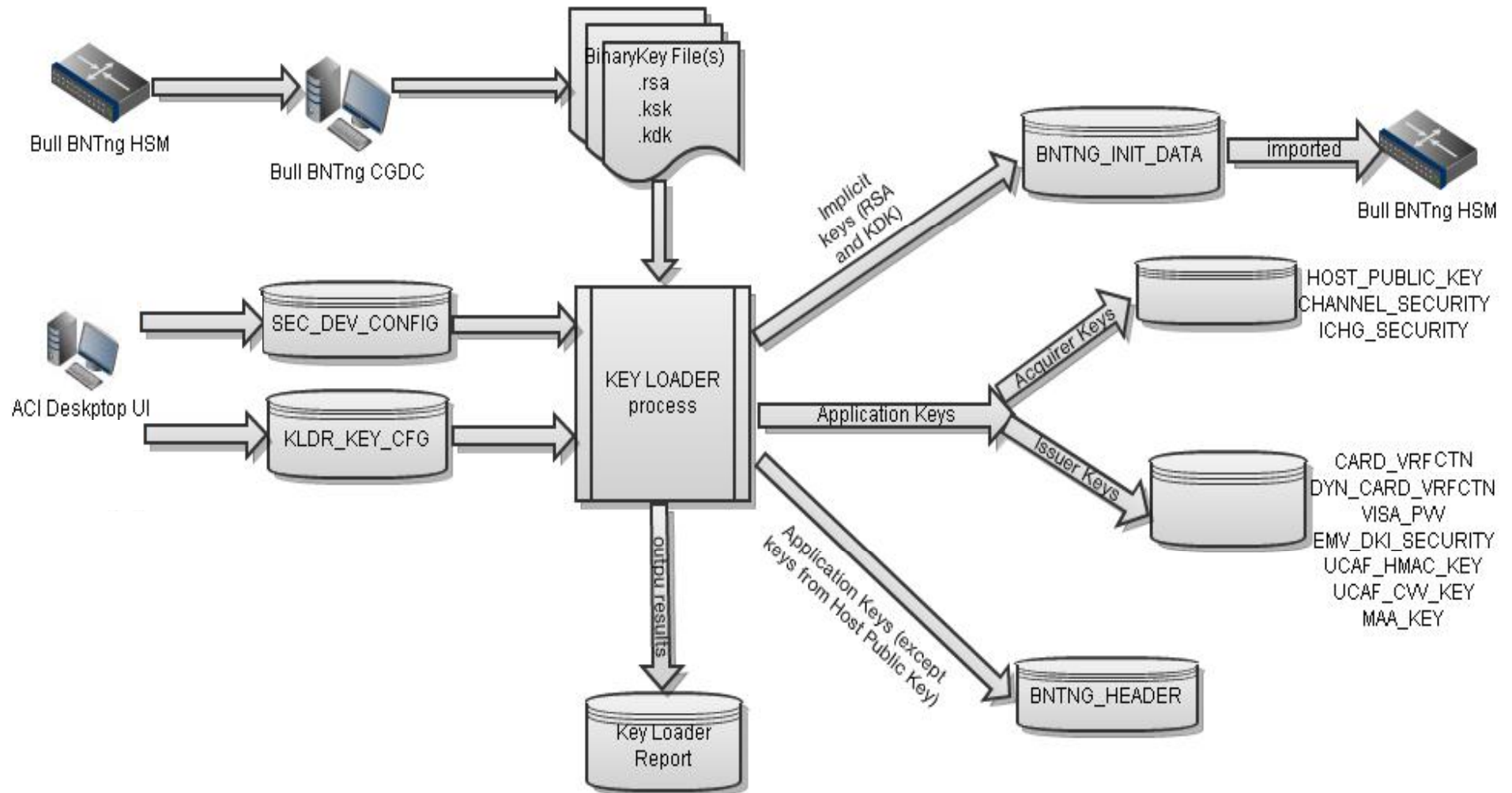
Transaction Security Configuration

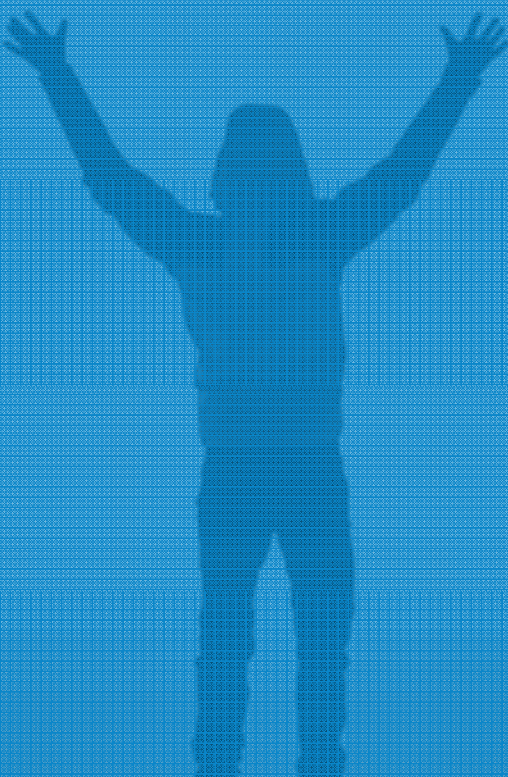
- Support for the French marketplace transaction security requirements using Bull BNTng HSM. Functionality includes:
 - AKDS NAC (Nouvelle Architecture Cryptographique)
 - expanded EMV Key Management (256 key sets)
 - Three Level Key Hierarchy for selected interfaces
 - key import process to load static keys



Transaction Security Configuration

- Key Loader Processing





TRANSACTION SECURITY

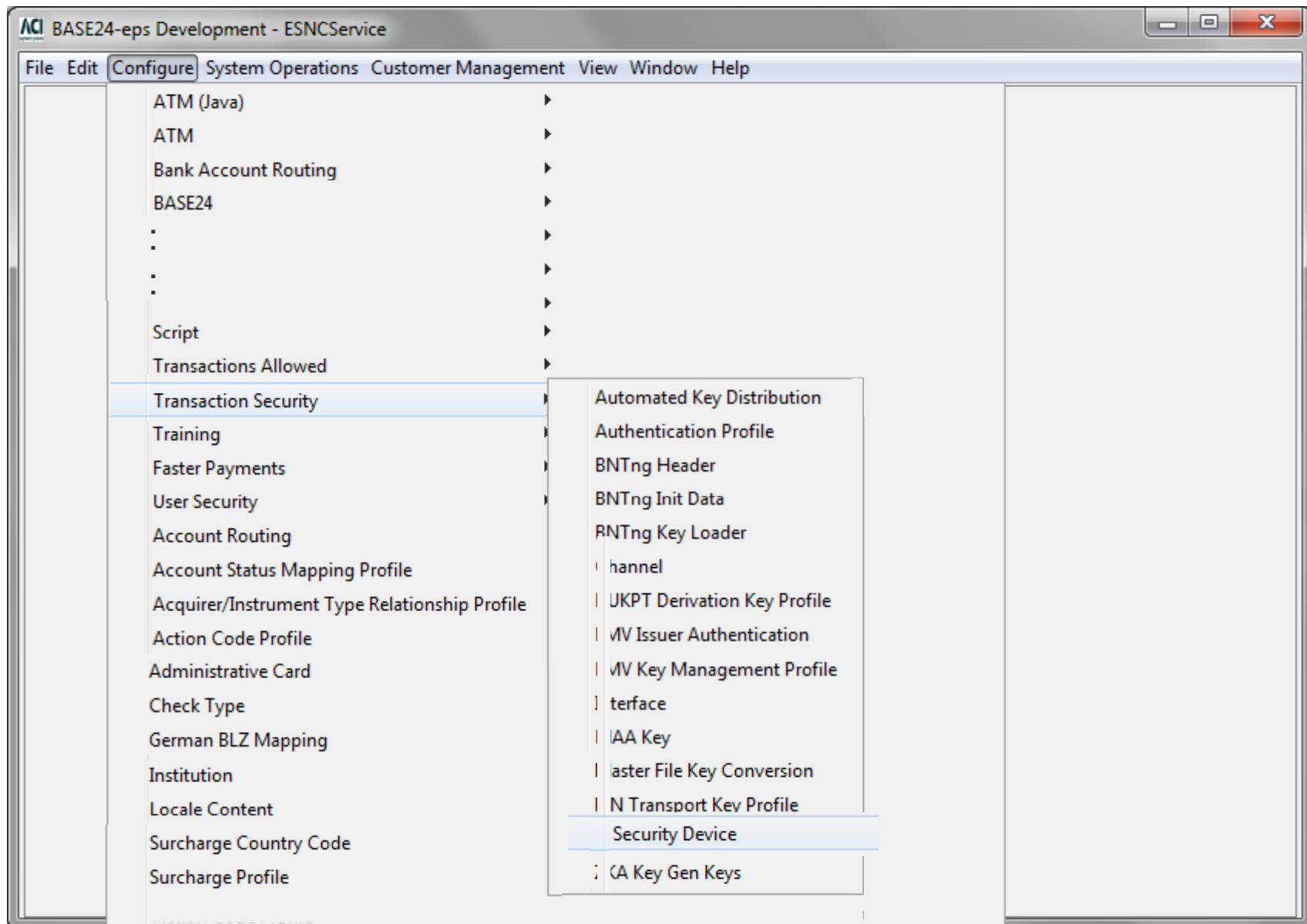
Transaction Security Configuration

Transaction Security Configuration

- Transaction security configuration includes configuration for:
 - Security Device
 - Channel
 - Customer Authentication (Issuer)
 - Authentication Profile
 - EMV Key Management Profile (optional)
 - Interface



Security Device



Security Device

The screenshot shows the 'Security Device Configuration - BOX1' window. It contains the following fields and values:

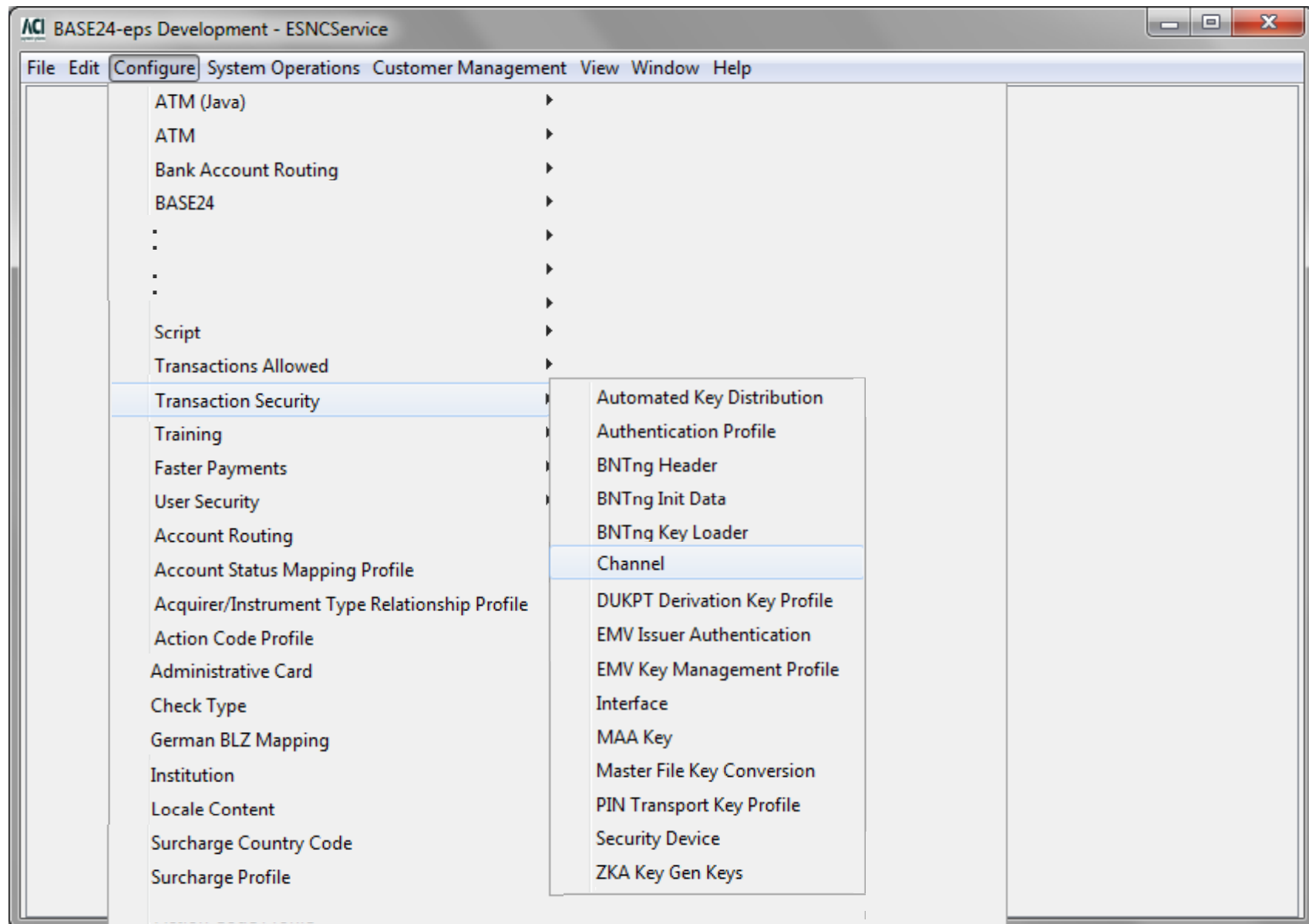
Field	Value
Security Device Name	BOX1
Security Device Type	Bull BNTng
Hardware Model	BNTng V7
Protocol/Interface	TCP/IP - ICE-XS UNIX
Key Scheme	N/A
Status	Active
Reset Timer Limit	60
Maximum Consecutive Errors	3
Error Notify Interval	10
Timeout Limit	5
Enable PKI	☒
Maximum Buffer Length	2048
Firmware Version	
HSM ID	0000
Configuration String	

At the bottom of the window, a status bar indicates: 2 - Successful view of BOX1.

Security device parameters and status for the HSM

The IBM CEX2 only requires one record to define the device type

Channel Security



Channel Security

EPS Development - ESNCService

File Edit Configure System Operations Customer Management View Window Help

Channel Security Configuration - 12345

Channel ID: 12345

External Key Block Format: Standard Atalla Old Key Timer Limit: 30

Key Hierarchy: Three Levels Exchange Key Option: Separate Key exchange keys

Data Encryption Exch Info | **Data Encryption Key Info** | Transaction Key Security | AS2805 Specific Keys | German Info

Export Master Info | PIN Information | MAC Information

Export Master Keys Key Length: Single

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current						
Future						

286 - Desktop bounds(0,0,1025,763)

Security configuration for ATM/POS devices. Used with Level 3 key hierarchy (currently no Device Handler components or device vendors support this level of encryption key).

Channel Security –Hierarchy

The screenshot shows the 'Key Hierarchy' dropdown set to 'Three Levels' and the 'Exchange Key Option' set to 'Single Key exchange key'. The 'Export Master Info' tab is highlighted with a red box. Below the tabs, the 'Export Master Keys' section is visible, showing a table with columns for Header, Key Part 1, Key Part 2, Key Part 3, MAC, and Check Digits. The 'Key Length' dropdown is set to 'Single'.

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current		0000 0000 0000 2B67				
Future		0000 0000 0000 56CE				

Selecting Three Level Key Hierarchy activates the Export Master Info tab.

The screenshot shows the 'Key Hierarchy' dropdown set to 'Two Levels' and the 'Exchange Key Option' set to 'Separate Key exchange keys'. The 'PIN Information' tab is highlighted with a red box. Below the tabs, the 'PIN Block Format' is set to 'ANSI PIN (ISO Format 0)' and the 'PIN Pad Character' is set to 'F'. The 'Exchange Keys' section is visible, showing a table with columns for Header, Key Part 1, Key Part 2, Key Part 3, MAC, and Check Digits. The 'Key Length' dropdown is set to 'Single'.

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current		1111 1111 1111 1111				
Future		1111 1111 1111 1111				

Selecting Two Level Key Hierarchy de-activates the Export Master Info tab.

Channel Security – PIN Information Tab

ACI EPS Development - ESNCService

File Edit Configure System Operations Customer Management View Window Help

ACI Channel Security Configuration - S1A^NDC

Channel ID: S1A^NDC

External Key Block Format: Standard Atalla Old Key Timer Limit: 30

Key Hierarchy: Two Levels Exchange Key Option: Single Key exchange key

Data Encryption Exch Info Data Encryption Key Info Transaction Key Security AS2805 Specific Keys German Info

Export Master Info PIN Information MAC Information

PIN Block Format: ANSI PIN (ISO Format 0) PIN Pad Character: F

Exchange Keys Key Variant: 0 - Key Exchange Key Variant Key Length: Single

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current		FCBA 7CF5 972C F0DD				
Future						

Inbound Keys Key Length: Single

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current						
Future						

286 - Desktop bounds(0,0,1025,763)

Future keys are only supported by FPS (Faster Payments) interface. They are stored in the CSECF data source.

Channel Security – MAC Information Tab

ACS EPS Development - ESNService

File Edit Configure System Operations Customer Management View Window Help

Channel Security Configuration - S1A^NDC

Channel ID: S1A^NDC

External Key Block Format: Standard Atalla Old Key Timer Limit: 60

Key Hierarchy: Two Levels Exchange Key Option: Single Key exchange key

Data Encryption Exch Info Data Encryption Key Info Transaction Key Security **AS2805 Specific Keys** German Info

Export Master Info PIN Information **MAC Information**

MAC Option: Single length DES Cipher Block Chaining MAC Length: 32 bits

Exchange/Inbound Outbound

Exchange Keys Export Variant: 0 - Key Exchange Key... Key Variant: 0 - Key Exchange Key... Key Length: Single

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current						
Future						

Inbound Keys Key Length: Single

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current						
Future						

286 - Desktop bounds(0,0,1025,763)

Channel Security – Data Encryption Exch Info

ACS Development - ESNService

File Edit Configure System Operations Customer Management View Window Help

Channel Security Configuration - S1A^NDC

Channel ID: S1A^NDC

External Key Block Format: Standard Atalla Old Key Timer Limit: 30

Key Hierarchy: Two Levels Exchange Key Option: Single Key exchange key

Export Master Info Data Encryption Exch Info Data Encryption Key Info PIN Information Transaction Key Security MAC Information AS2805 Specific Keys German Info

Exchange Keys Key Variant: 0 - Key Exchange Key Variant Key Length: Single

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current						
Future						

Initialization Vector

	Header	Key Part 1	Key Part 2	Key Part 3	MAC
1					

Used with Interac switch to support encrypted balance data (Exchange key tab)

286 - Desktop bounds(0,0,1025,763)

Channel Security – Data Encryption Key Tab

The screenshot shows the 'Channel Security Configuration - S1A^NDC' window. The 'Data Encryption Key Info' tab is selected. The configuration includes fields for Channel ID (S1A^NDC), External Key Block Format (Standard Atalla), Key Hierarchy (Two Levels), Old Key Timer Limit (30), and Exchange Key Option (Single Key exchange key). Below these are tabs for Export Master Info, PIN Information, and MAC Information. The 'Data Encryption Key Info' tab contains fields for DES Mode (ECB), Data Type (Unpacked ASCII hexadecimal), and Encryption Layers (1). The 'Inbound' sub-tab is active, showing a table for Inbound Keys with columns for Header, Key Part 1, Key Part 2, Key Part 3, MAC, and Check Digits. The table has four rows: Current, Future 1, and Future 2. The 'Key Length' is set to Single.

Channel ID: S1A^NDC

External Key Block Format: Standard Atalla

Old Key Timer Limit: 30

Key Hierarchy: Two Levels

Exchange Key Option: Single Key exchange key

DES Mode: ECB

Data Type: Unpacked ASCII hexadecimal

Encryption Layers: 1

Inbound Keys

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current...						
Current...						
Future 1						
Future 2						

Key Length: Single

Used with Interac Interchange (Canada) to support encrypted balance data (Encryption key tab) at POS devices

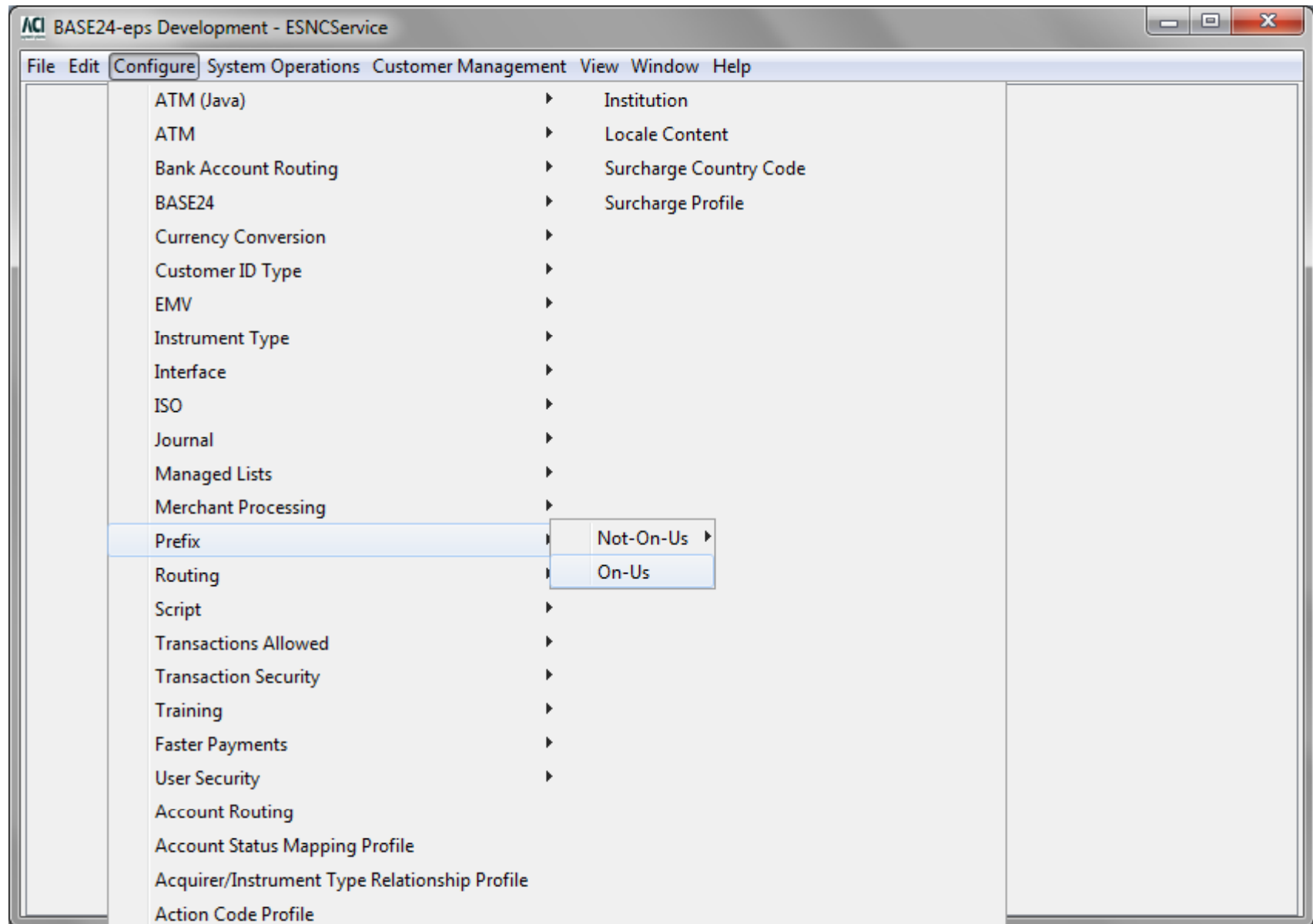
Channel Configuration/Channel Security

The image displays three overlapping software windows from the ACI system, each featuring a 'Channel ID' field highlighted with a red box. The windows are:

- ATM Channel Configuration:** Shows fields for 'Channel ID', 'Institution ID', 'Device Type', and 'ATM Channel Profile'. It includes tabs for 'Receipt Profiles', 'Other Profiles', 'Load Groups', 'Migration', 'Migration (Locale)', and 'Security'.
- Merchant Channel Configuration:** Shows fields for 'Channel ID', 'Channel Name', 'Institution ID', and 'Merchant Name'. It includes tabs for 'Key Management', 'Clerk', 'Accumulators', and 'Instrument'.
- Channel Security Configuration:** Shows fields for 'Channel ID', 'External Key Block Format', 'Key Hierarchy', 'PIN Block Format', and 'PIN Pad Character'. It includes tabs for 'Data Encryption Exch Info', 'Data Encryption Key Info', 'Transaction Key Security', 'AS2805 Specific Keys', and 'German Info'.

Channel ID: 16 character alphanumeric

Issuer – Customer Authentication



Issuer – Customer Authentication

ACI EPS Development - ESNCService

File Edit Configure System Operations Customer Management View Window Help

ACI On-Us Prefix Configuration

Prefix: Select/Enter PAN Length: Select/Enter

Issuer ID:

Card Verification Dynamic Card Verification EMV General Options **EMV Auth Options** CAVV

Processing Information Check Processing Issuer Information **Card Track Information** PIN Verification

Track Processing Sequence

Primary: Track 2 Secondary: Track 1 Tertiary: Track 3

Track 1 Settings Track 2 Settings Track 3 Settings

Track Length

Maximum: 0 Minimum: 0

Card Verification: 3 Dynamic Card Verification: 0 Application Tran Counter: 0

Location on Track

Member Number: 0 Service Code: 0 Dynamic Card Verification: 0

PVV Key Index: 0 Expiration Date: 0 Application Tran Counter: 0

PVV: 0 Card Verification: 0 Unpredictable Number: 0

286 - Desktop bounds(0,0,1025,763)

Security
parameters
related to the card
track

Issuer – Customer Authentication

ACI EPS Development - ESNCService

File Edit Configure System Operations Customer Management View Window Help

On-Us Prefix Configuration

Prefix: Select/Enter PAN Length: Select/Enter

Issuer ID:

Card Verification Dynamic Card Verification EMV General Options EMV Authentication Options CAVV

Processing Information Check Processing Issuer Information Card Track Information PIN Verification

PIN Location

☐ No PIN Verification Number available

☐ PIN Verification Number on Cardholder File

☐ PIN Verification Number on Card Track

PIN Verification Type: IBM DES 3624

PIN Verification Number Length: 4

PIN Verification Profile: AUT1, AUT2, AUT3, AUT4, AUT7

286 - Desktop bounds(0,0,1025,763)

Security parameters related to PIN verification

Issuer – Customer Authentication

ACI EPS Development - ESNCService

File Edit Configure System Operations Customer Management View Window Help

ACI On-Us Prefix Configuration

Prefix: Select/Enter PAN Length: Select/Enter

Issuer ID:

Processing Information | Check Processing | Issuer Information | Card Track Information | PIN Verification

Card Verification | Dynamic Card Verification | EMV General Options | EMV Auth Options | CAVV

Card Verification Type:

Card Verification Profile:

Effective Date: August 2011 Manual Date Check:

Card Verification Retry: ☐

Manual Card Verification Type: No CVD2 required

Manual Card Entry Option: Do not deny manually entered transactions

Card Verification Failure Action:

Complete Track:

Uncertain Track:

Continue Authorization
Decline - Return Card
Decline - Retain Card
Referral

286 - Desktop bounds(0,0,1025,763)

Security parameters related to card verification

Issuer – Customer Authentication

ACI EPS Development - ESNCService

File Edit Configure System Operations Customer Management View Window Help

ACI On-Us Prefix Configuration

Prefix: Select/Enter PAN Length: Select/Enter

Issuer ID:

Processing Information Check Processing Information Issue Information Card Track Information PIN Verification

Card Verification Dynamic Card Verification EMV General Options EMV Auth Options CAVV

Dynamic Card Verification Type: Do not check Dynamic Verification Card Digits

Dynamic Card Verification Profile:

Pre-screen Dynamic Card Verification Check: ☐

Bad Dynamic Card Verification Action: Deny and return the card

286 - Desktop bounds(0,0,1025,763)

Security parameters related to Dynamic Card Verification (RFID: Visa's Wave and MasterCard PayPass)
Also known as Proximity cards

Issuer – Customer Authentication

ACI BASE24-eps Development - ESNCSERVICE

File Edit Configure System Operations Customer Management View Window Help

On-Us Prefix Configuration

Prefix: Select/Enter PAN Length: Select/Enter

Issuer ID: Select/Enter

Processing Information		Check Processing		Issuer Information		Card Track Information	
PIN Verification	Card Verification	Dynamic Card Verification	EMV General Options	EMV Auth Options	CAVV		
Primary EMV Issuer Application Data Format		None					
Alternate EMV Issuer Application Data Format		None					
Card Authentication Type		Do not perform CAM check					
EMV DKI Security		☐		EMV Profile		Select/Enter	
Transport Key Locator		Select/Enter		EMV Script MAC Length		4	
EMV Destination Routing Profile		Select/Enter		EMV Card Verification Results Profile		Select/Enter	
EMV Terminal Verification Results Profile		Select/Enter		EMV Reason Online Code Profile		Select/Enter	
EMV Card Verification		Offline PIN Support					
EMV Check Type		Do not perform card verification		Effective Date		January 2005	
EMV Check Method		Use non-EMV card verification		EMV Service Code		000	
Bad PIN Tries Action		Do not perform EMV specific processing					
PIN Management		Do not perform EMV specific processing					
☐ Use PIN Support Date		July 2013					

Security parameters related to EMV processing

Issuer – Customer Authentication

The screenshot displays the 'On-Us Prefix Configuration' window within the 'EPS Development - ESNCServices' application. The window features a menu bar (File, Edit, Configure, System Operations, Customer Management, View, Window, Help) and a toolbar. The main configuration area is divided into several tabs: Processing Information, Check Processing, Issuer Information, Card Track Information, PIN verification, Card Verification, Dynamic Card Verification, EMV General Options, EMV Auth Options (highlighted with a red box), and CAVV. The 'EMV Auth Options' tab is active, showing various security parameters. Under 'Pre-screening Options', there are checkboxes for Card Authentication Verification, Application Transaction Counter Verification, Card Verification Results, Terminal Verification Results, Reason Online Code Verification, and Fallback Check. To the right, there are checkboxes for Card Verification Results, Terminal Verification Results, Reason Online Code Verification, and Force Fallback to Over-the-Floor Indicator. Below these, there is a section for 'Application Transaction Counter Verification' with a dropdown menu set to 'Do not perform ATC'. The 'Next Online Date' is set to '30'. The 'Card Authentication Failure Action' section includes dropdowns for 'Reliable' (Continue Authorization) and 'Unreliable' (Continue Authorization). The 'Fallback Action' is set to 'Continue processing the transact...' and the 'ATC Bad Disposition Action' is set to 'Deny and return the card'.

Security parameters related to EMV processing

Issuer – Customer Authentication

The screenshot shows the 'On-Us Prefix Configuration' window within the 'EPS Development - ESNCServices' application. The window has a menu bar (File, Edit, Configure, System Operations, Customer Management, View, Window, Help) and a toolbar. The 'On-Us Prefix Configuration' sub-window contains several fields and tabs. The 'CAVV' tab is selected, showing options for 'Secure Internet Validation Check Type' (set to 'Secure Internet Validation disabled'), 'CAVV Profile' (highlighted with a red box), 'Secure Internet Validation Pre-screening Check' (set to 'Do not verify CAVV during pre-screening'), and 'CAVV Authentication Failure Action' (with sub-options for 'CAVV Validation Check Failure Action' and 'CAVV Not Present Action', both set to 'Note the situation and continue').

Prefix: PAN Length:
Issuer ID:

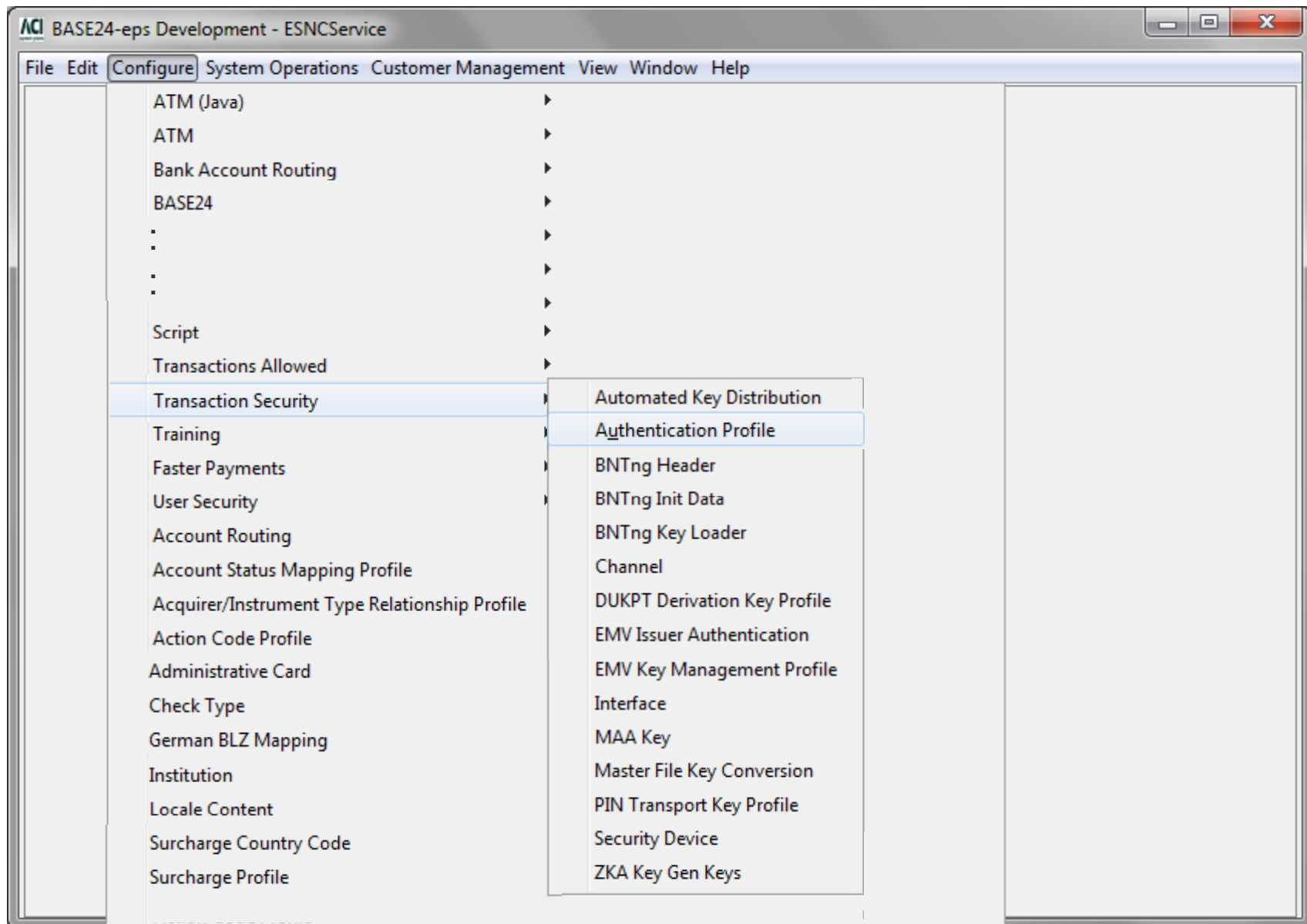
Processing Information | Check Processing | Issuer Information | Card Track Information | PIN Verification
Card Verification | Dynamic Card Verification | EMV General Options | EMV Auth Options | CAVV

Secure Internet Validation Check Type:
CAVV Profile:
Secure Internet Validation Pre-screening Check:
CAVV Authentication Failure Action
CAVV Validation Check Failure Action:
CAVV Not Present Action:

286 - Desktop bounds(0,0,1025,763)

Security parameters related to Secure Internet Validation (CAVV-Cardholder Authentication Verification Value). Verified by Visa

Authentication Profile



Authentication Profile

ACI Desktop - ESNCService

File Edit Configure System Operations Customer Management View Window Help

ACI Authentication Profile Configuration - IBM_DES

Profile: IBM_DES

Begin Date	End Date	IBM DES	Visa PVV	Diebold	NCR	Identikey	EMV	Card	UCAF
April 2014	April 2016	☒	None	☐	☐	☐	EMV	Both...	None

Chip Auth	Card	Contactless	Dual PVN/PIN	UCAF HMAC	UCAF CVV		
			Visa PVV Keys	Diebold	NCR	Identikey	EMV

DES (IBM 3624) PIN Verification

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Ch
1	1V3NE000	1111 2222 3333 4444	2222 3333 4444 5555	3333 4444 5555 6666	4444 5555 6666 7777	

Decimalization Table: 0123456789123456

PAN Pad Character: F

PAN Verify Length: 4

PAN Verify Offset: 0

4 - Successful view of Profile IBM_DES April 2014 - April 2016.

190 - Security Information - Last successful login: 6/23/14 3:59 PM

Keys and data required for the IBM 3624 DES verification method

Authentication Profile

The screenshot shows the 'Authentication Profile Configuration - IBM_DES' window. The 'Profile' dropdown is set to 'IBM_DES'. Below it, there are tabs for 'Begin Date', 'End Date', 'IBM DES', 'Visa PVV', 'Diebold', 'NCR', 'Identikey', 'EMV', 'Card', and 'UCAF'. The 'Visa PVV' tab is selected, showing a 'Both...' dropdown. Below this, there are tabs for 'Chip Auth', 'Card', 'Contactless', 'Dual PIN PIN', 'UCAF HMAC', and 'UCAF CVV'. The 'Dual PIN PIN' tab is selected, showing a 'Visa PVV Keys' tab. Below this, there is a table with columns: 'Header', 'Key Part 1', 'Key Part 2', 'Key Part 3', and 'MAC'. The table contains three rows of data, each with a unique header and key parts. The status bar at the bottom indicates '190 - Security Information - Last successful login: 6/23/14 3:59 PM'.

Keys and data required for the VISA PVV verification method.

Additional key pairs are made available for the Bull HSM.

	Header	Key Part 1	Key Part 2	Key Part 3	MAC
1	IMKSMI02	6A51 F994 E458 4232	1D1F 81BA 1CC3 B43F	33F2 2166 CC44 51BA	E506 68FF 6622 1234
2	IMKSMI02	6A51 F994 E458 4232	1D1F 81BA 1CC3 B43F	33F2 2166 CC44 51BA	E506 68FF 6622 1234
3	IMKSMI02	6A51 F994 E458 4232	1D1F 81BA 1CC3 B43F	33F2 2166 CC44 51BA	E506 68FF 6622 1234
4					
5					
6					

Authentication Profile

ACI Desktop - ESNCService

File Edit Configure System Operations Customer Management View Window Help

ACI Authentication Profile Configuration - IBM_DES

Profile: IBM_DES

Begin Date	End Date	IBM DES	Visa PVV	Diebold	NCR	Identikey	EMV	Card	UCAF
April 2014	April 2016	☒	Both ...	☒	☐	☐	EMV	Both...	None

Chip Auth Card Contactless Dual PVN PIN UCAF HMAC UCAF CVV

DES (IBM 3624) PIN Verification Visa PVV Keys **Diebold** NCR Identikey EMV

Relative Location: 1 Algorithm Number: Entry Type: Clear

Diebold Number Table

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Ch
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

190 - Security Information - Last successful login: 6/23/14 3:59 PM

Keys and data
required for the
Diebold
verification
method

Authentication Profile

ACI Desktop - ESNService

File Edit Configure System Operations Customer Management View Window Help

ACI Authentication Profile Configuration - IBM_DES

Profile: IBM_DES

Begin Date	End Date	IBM DES	Visa PVV	Diebold	NCR	Identikey	EMV	Card	UCAF
April 2014	April 2016	☐	Both ...	☒	☐	☒	EMV	Both...	None

Chip Auth	Card	Contactless	Dual PVN PIN	UCAF HMAC	UCAF PVV
DES (IBM 3624) PIN Verification	Visa PVV Keys	Diebold	NCR	Identikey	EMV

Bank ID:

PAN Verify Offset:

PAN Verify Length:

Compare Indicator:

190 - Security Information - Last successful login: 6/23/14 3:59 PM

Keys and data required for the Identikey verification method

Authentication Profile

ACI BASE24-eps Development - ESNCSERVICE

File Edit Configure System Operations Customer Management View Window Help

ACI Authentication Profile Configuration - AAA_CVV_01

Profile: AAA_CVV_01

Begin Date	End Date	IBM DES	Visa PVV	Diebold	NCR	Identikey	EMV	Card	UCAF
July 2013	July 2013	☐	None	☐	☐	☐	Bot...	Both...	UCAF...

Chip Auth	Card	Contactless	Dual PVN PIN	UCAF HMAC	UCAF CVV
DES (IBM 3624) PIN Verification	Visa PVV Keys	Diebold	NCR	Identikey	EMV

EMV Scheme: Visa

Branch/Height Parameters: Branch factor 2; Tree Height...

Cryptogram Version Number: 0E

Destination PIN Block Type: Visa Format Without using...

Secure Message Key for Integrity

	Header	Key Part 1	Key Part 2	Key Part 3	Check Digits
1					

Secure Message Key for Confidentiality

	Header	Key Part 1	Key Part 2	Key Part 3	Check Digits
1					

Cryptographic Keys

	Key Index	Header	Key Part 1	Key Part 2	Key Part 3	Check Digits
1						
2						

Keys and data required for EMV chip card verification

Authentication Profile

ACI BASE24-eps Development - ESNCSERVICE

File Edit Configure System Operations Customer Management View Window Help

ACI Authentication Profile Configuration - AAA_CVV_01

Profile: AAA_CVV_01

Begin Date	End Date	IBM DES	Visa PVV	Diebold	NCR	Identikey	EMV	Card	UCAF
July 2013	July 2013	☐	None	☐	☐	☐	Bot...	Both...	UCAF...

DES (IBM 3624) PIN Verification Visa PVV Keys Diebold NCR Identikey EMV

Chip Auth Card Contactless Dual PIN UCAF HMAC UCAF CVV

EMV Cryptography

EMV Key Profile: AUT1 EMV Issuer Application Data Format: VISA

Card Configuration

Application Interchange Profile: Predicted Card Verification Results:

Use of Initial Card Verification Results: Never Initial Card Verification Results:

Chip Authentication Configuration

Application: CAP

Internet Authentication Flags: Issuer Proprietary Bitmap:

Number of Resynch Attempts: 0 HSM Specific Parameters:

EMV
configuration for
Chip
Authentication
Program (CAP)

Authentication Profile

ACI BASE24-eps Development - ESNCSERVICE

File Edit Configure System Operations Customer Management View Window Help

ACI Authentication Profile Configuration - AAA_CVV_01

Profile AAA_CVV_01

Begin Date	End Date	IBM DES	Visa PVV	Diebold	NCR	Identikey	EMV	Card	UCAF
July 2013	July 2013	☐	None	☐	☐	☐	None	Both...	UCAF...

DES (IBM 3624) PIN Verification	Visa PVV Keys	Diebold	NCR	Identikey	EMV
Chip Auth	Card	Contactless	Dual PVN PIN	UCAF HMAC	UCAF CVV
1					

Keys required for card verification

Authentication Profile

ACI BASE24-eps Development - ESNCSERVICE

File Edit Configure System Operations Customer Management View Window Help

ACI Authentication Profile Configuration - AAA_CVV_01

Profile AAA_CVV_01

Begin Date	End Date	IBM DES	Visa PVV	Diebold	NCR	Identikey	EMV	Card	UCAF
July 2013	July 2013	☐	None	☐	☐	☐	None	Both...	UCAF...

DES (IBM 3624) PIN Verification Visa PVV Keys Diebold NCR Identikey EMV

Chip Auth Card **Contactless** Dual PVN PIN UCAF HMAC UCAF CVV

Contactless Card Verification Algorithm Visa

Header	Key Part 1	Key Part 2	Key Part 3	Check Digits
1				

Keys required for Contactless Magnetic Stripe support

Authentication Profile

ACI BASE24-eps Development - ESNCSERVICE

File Edit Configure System Operations Customer Management View Window Help

ACI Authentication Profile Configuration - AAA_HMAC_01

Profile: AAA_HMAC_01

	Begin Date	End Date	IBM DES	Visa PVV	Diebold	NCR	Identikey	EMV	Card	UCAF
	July 2013	July 2013	☐	None	☐	☐	☐	None	None	UCAF...

DES (IBM 3624) PIN Verification			Visa PVV Keys		Diebold	NCR	Identikey	EMV
Chip Auth	Card	Contactless	Dual PVN PIN		UCAF HMAC		UCAF CVV	

ACS ID: Select BIN Key ID: Select

	Header	Key	Check Digits
1			

Keys required for UCAF HMAC support

Authentication Profile

ACI BASE24-eps Development - ESNCSERVICE

File Edit Configure System Operations Customer Management View Window Help

ACI Authentication Profile Configuration - AAA_CVV_01

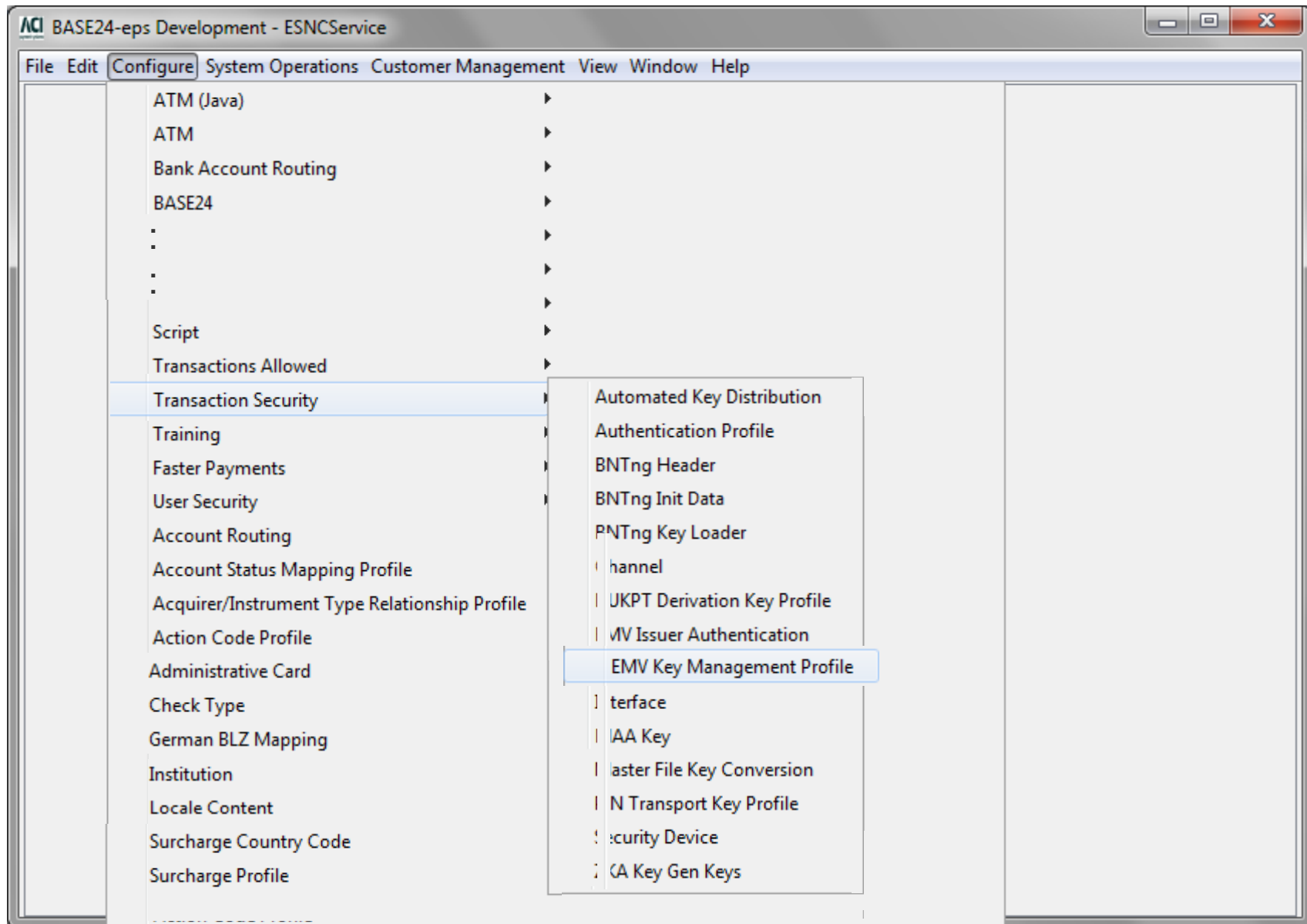
Profile AAA_CVV_01

Begin Date	End Date	IBM DES	Visa PVV	Diebold	NCR	Identikey	EMV	Card	UCAF
July 2013	July 2013	☐	None	☐	☐	☐	None	None	UCAF...

DES (IBM 3624) PIN Verification			Visa PVV Keys		Diebold	NCR	Identikey	EMV																		
Chip Auth	Card	Contactless	Dual PVN PIN		UCAF HMAC		UCAF CVV																			
ACS ID Select BIN Key ID Select <table border="1"><thead><tr><th></th><th>Header</th><th>Key Part 1</th><th>Key Part 2</th><th>Key Part 3</th><th>Check Digits</th></tr></thead><tbody><tr><td>1</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr></tbody></table>										Header	Key Part 1	Key Part 2	Key Part 3	Check Digits	1											
	Header	Key Part 1	Key Part 2	Key Part 3	Check Digits																					
1																										

Keys required for UCAF CVV support

EMV Key Management Profile



EMV Key Management Profile

ACI BASE24-eps Development - ESNCSERVICE

File Edit Configure System Operations Customer Management View Window Help

ACI EMV Key Management Profile Configuration - DCA_TEST_001

EMV Profile: DCA_TEST_001

	Begin Date	End Date	Derivation Key Index (DKI)		
			MSB	LSB	Decimal
	April 2013	April 2013	2	4	36
	April 2013	April 2013	3	3	51
	April 2013	April 2013	4	4	68

EMV Scheme: Visa Branch Height Parameters: Branch factor 2; Tree Height 16

Cryptographic Version Number: 0E Destination PIN Block Type: Visa Format Without using Curre...

Secure Message Key For Integrity

	Header	Key Part 1	Key Part 2	Key Part 3	Check Digits
1	IMKSMI02	6A51 F994 E458 4232	1D1F 81BA 1CC3 B43F		E50668

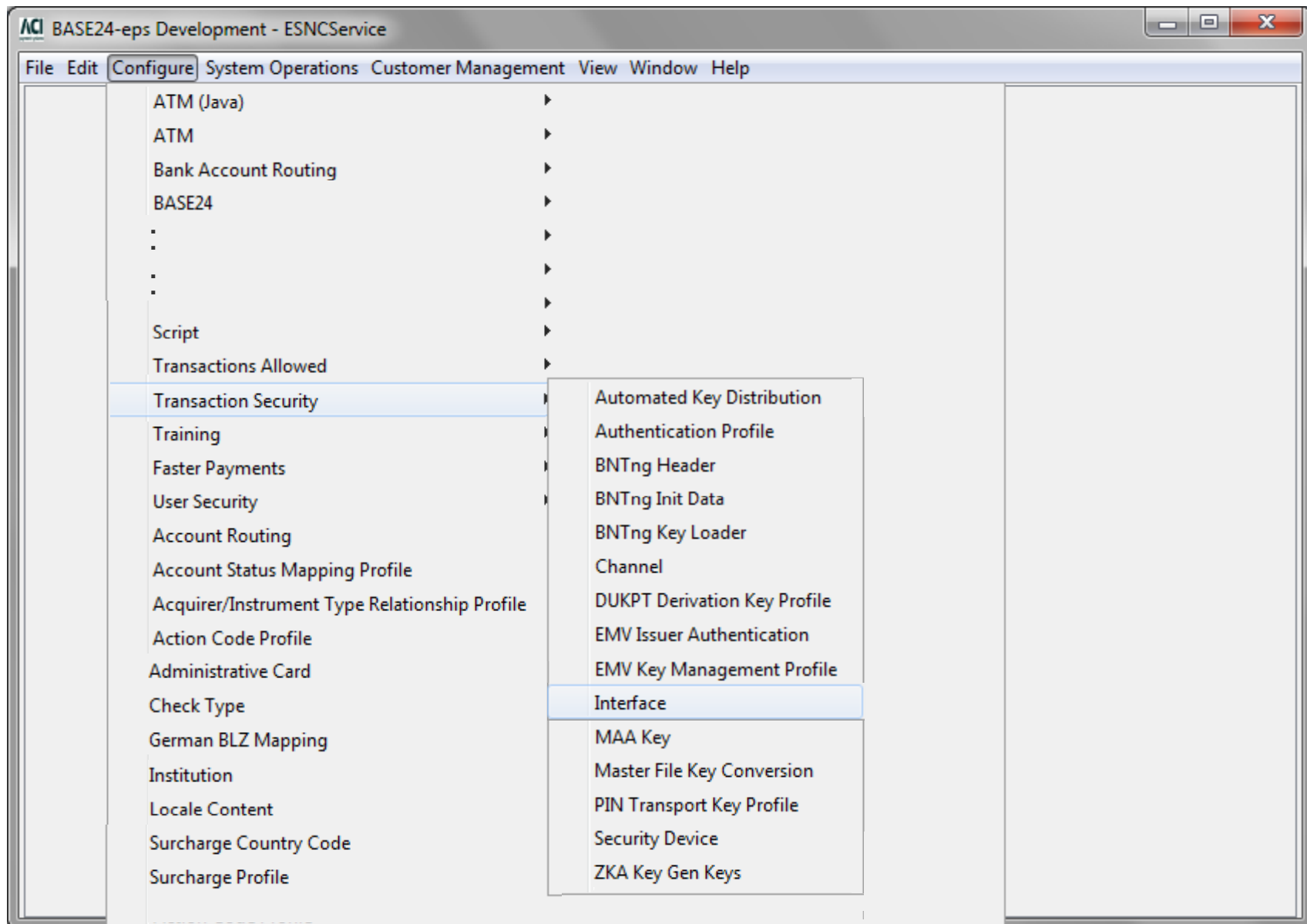
Secure Message Key For Confidentiality

	Header	Key Part 1	Key Part 2	Key Part 3	Check Digits
1	IMKSMI02	6A51 F994 E458 4232	1D1F 81BA 1CC3 B43F		E50668

Application Cryptogram Key

	Header	Key Part 1	Key Part 2	Key Part 3	Check Digits
1	IMKSMI02	6A51 F994 E458 4232	1D1F 81BA 1CC3 B43F		E50668

Interface Security



Interface Security

Series keys are currently supported by e-RSB (France) Interface.

Supports 3 level hierarchy, enter Transport Master Key

Also supports future keys

Selecting **Key Hierarchy - Three Level** activates the Master Info tab and Level 2 Import Master Info tab. Used by Interac Association's PIN security requirements (Canada), Faster Payments (FPS) security requirements (Europe), e-RSB Interface (France)

keys.

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current						
Future						

Interface Security

ACS Development - ESNService

File Edit Configure System Operations Customer Management View Window Help

Interface Security Configuration

Interface Name: SELECT/ENTER Series: SELECT/ENTER

Use Series: ☐

External Key Block Format: Standard Atalla Old Key Timer Limit: 30

Key Hierarchy: Three Levels Exchange Key Option: Single Key exchange key

MAC Information Data Entry Exchange Info Data Entry Key Info German Info Master Keys AS2805 Specific Keys

Master Info Level 2 Import Master Info Exchange Information PIN Information

Import Master PIN Keys Import Master MAC Keys Import Master Data Keys

PIN Keys Key Length: Single

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current						
Future						

286 - Desktop bounds(0,0,1025,763)

Import Master Info for PIN Keys, MAC Keys, Data Keys where separate master keys are required.

Supported by Interac Association's PIN security requirements (Canada)

Faster Payments (FPS) security requirements (Europe)

e-RSB Interface (France)

Interface Security

ACS EPS Development - ESNCService

File Edit Configure System Operations Customer Management View Window Help

Interface Security Configuration

Interface Name: SELECT/ENTER Series: SELECT/ENTER

Use Series: ☐

External Key Block Format: Standard Atalla Old Key Timer Limit: 30

Key Hierarchy: Three Levels Exchange Key Option: Single Key exchange key

MAC Information Data Encr Exch Info Data Encr Key Info German Info Master Keys AS2805 Specific Keys

Master Info Level 2 Import Master Info Exchange Information PIN Information

Keys Generated Online ☐

PIN Keys MAC Keys

PIN Keys Option: Use import key for both functions Key Variant: 0 - Key Exchange Ke... Key Length: Single

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Import...						
Export(...						
Import...						
Export(F)						

286 - Desktop bounds(0,0,1025,763)

KEK – used for dynamic key exchange of PIN or MAC encryption keys.

Interface Security

ACI EPS Development - ESNCService

File Edit Configure System Operations Customer Management View Window Help

Interface Security Configuration

Interface Name Series

Use Series ☐

External Key Block Format Old Key Timer Limit

Key Hierarchy Exchange Key Option

MAC Information Data Encr Exch Info Data Encr Key Info German Info Master Keys AC2005 Specific Keys

Master Info Level 2 Import Master Info Exchange Information **PIN Information**

PIN Block Format PIN PAD Character

Outbound Keys Key Length Current Index Key Counter 0

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current...						
Future						

Inbound Keys Key Length Current Index Key Counter 0

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current...						
Future						

286 - Desktop bounds(0,0,1025,763)

Interface Security

ACS Development - ESNCService

File Edit Configure System Operations Customer Management View Window Help

Interface Security Configuration

Interface Name: SELECT/ENTER Series: SELECT/ENTER

Use Series: ☐

External Key Block Format: Standard Atalla Old Key Timer Limit: 30

Key Hierarchy: Three Levels Exchange Key Option: Single Key exchange key

Master Info
MAC Information
Level 2 Import Master Info
Data Encr Exch Info
Data Encr Key Info
Exchange Information
German Info
Master Keys
PIN Information
AS2805 Specific Keys

MAC Option: Single length DES Cipher Block Chaining MAC Length: 32 bits

Outbound Keys Key Length: Single Current Index: 1 Key Counter 0

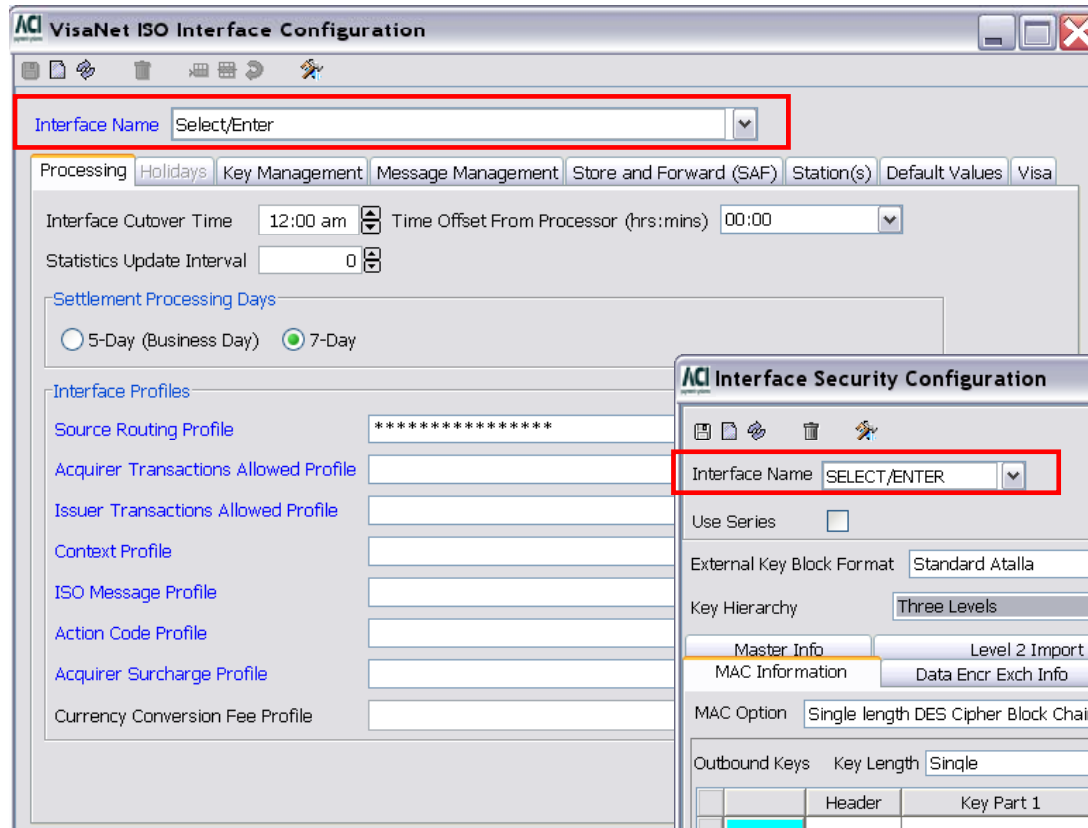
	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current...						
Future						

Inbound Keys Key Length: Single Current Index: 1 Key Counter 0

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current...						
Future						

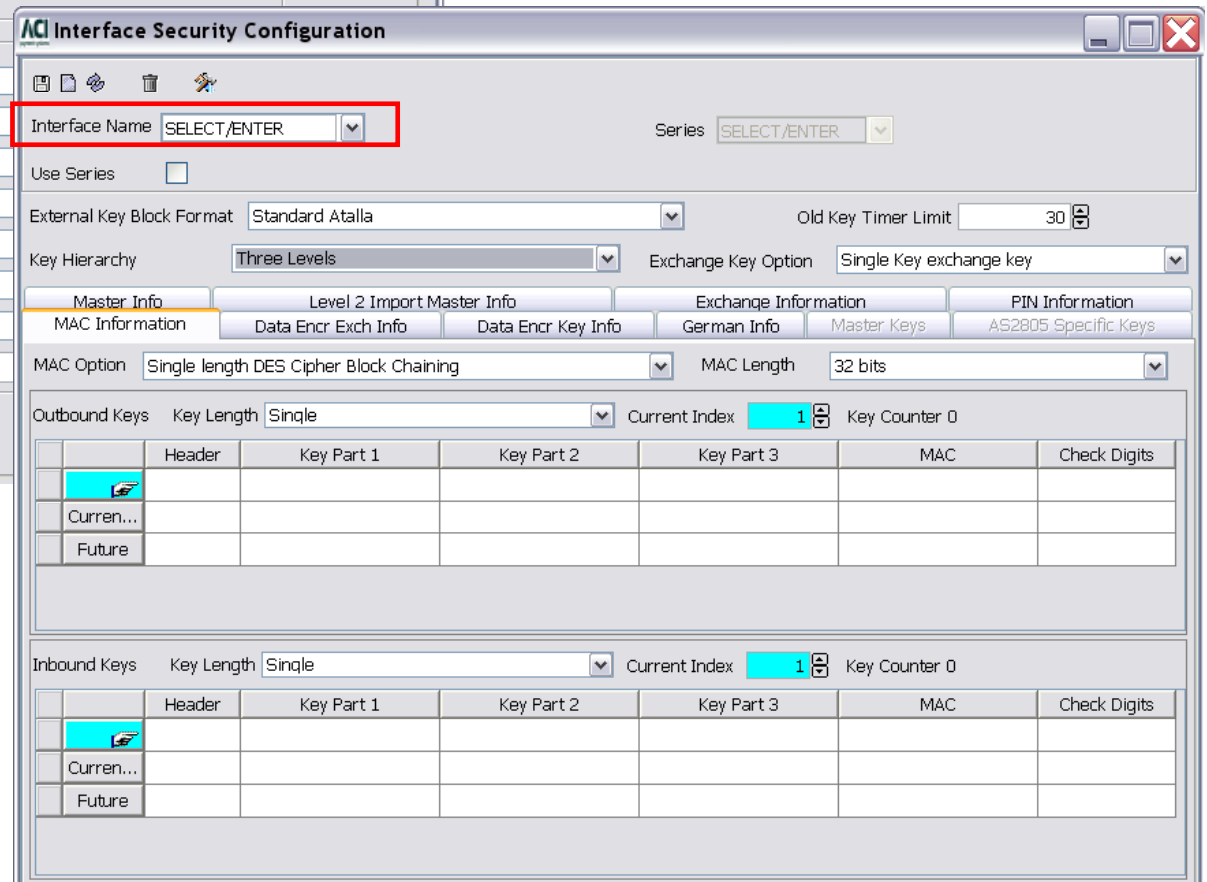
286 - Desktop bounds(0,0,1025,763)

Interface Configuration/Interface Security



The screenshot shows the 'VisaNet ISO Interface Configuration' window. The 'Interface Name' field is highlighted with a red box and contains the text 'Select/Enter'. Below this, there are tabs for 'Processing', 'Holidays', 'Key Management', 'Message Management', 'Store and Forward (SAF)', 'Station(s)', 'Default Values', and 'Visa'. The 'Processing' tab is active, showing fields for 'Interface Cutover Time' (12:00 am), 'Time Offset From Processor (hrs:mins)' (00:00), and 'Statistics Update Interval' (0). There are also radio buttons for 'Settlement Processing Days' (5-Day (Business Day) and 7-Day). A list of 'Interface Profiles' is shown on the left, including Source Routing Profile, Acquirer Transactions Allowed Profile, Issuer Transactions Allowed Profile, Context Profile, ISO Message Profile, Action Code Profile, Acquirer Surcharge Profile, and Currency Conversion Fee Profile.

The Interface Name field will match.



The screenshot shows the 'Interface Security Configuration' window. The 'Interface Name' field is highlighted with a red box and contains the text 'SELECT/ENTER'. To its right is a 'Series' field with 'SELECT/ENTER'. Below these are fields for 'Use Series' (checkbox), 'External Key Block Format' (Standard Atalla), 'Old Key Timer Limit' (30), 'Key Hierarchy' (Three Levels), and 'Exchange Key Option' (Single Key exchange key). There are tabs for 'Master Info', 'Level 2 Import Master Info', 'Exchange Information', and 'PIN Information'. The 'Master Info' tab is active, showing 'MAC Information' and 'Data Encr Exch Info'. Below this are fields for 'MAC Option' (Single length DES Cipher Block Chaining) and 'MAC Length' (32 bits). There are sections for 'Outbound Keys' and 'Inbound Keys', each with a table for key management. The 'Outbound Keys' section has a 'Current Index' of 1 and a 'Key Counter' of 0. The 'Inbound Keys' section also has a 'Current Index' of 1 and a 'Key Counter' of 0. Both sections have a table with columns for Header, Key Part 1, Key Part 2, Key Part 3, MAC, and Check Digits. The 'Future' row in both tables is highlighted with a blue background.

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current...						
Future						

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Check Digits
Current...						
Future						

Future keys are only supported by Faster Payments Service (FPS) interface

Interface Configuration/Interface Security

ACI Interface Security Configuration

Interface Name: SELECT/ENTER

Series: 1 (BASE) Status: Enabled

Use Series: ☒

External Key Block Format: Standard Atalla Old Key Timer Limit: 30

Key Hierarchy: Two Levels Exchange Key Option: Single Key exchange key

Data Encr Exch Info | Data Encr Key Info | German Info | Master Keys | AS2805 Specific Keys

Master Info | Level 2 Import Master Info | Exchange Information | PIN Information | MAC Information

PIN Block Format: ANSI PIN (ISO Format 0) PIN PAD Character: F

Outbound Keys Key Length: Single Current Index: 1 Key Counter: 0

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Ch
Current...						
Future						

Inbound Keys Key Length: Single Current Index: 1 Key Counter: 0

	Header	Key Part 1	Key Part 2	Key Part 3	MAC	Ch
Current...						
Future						

Defines multiple sets (or series) of keys for a single interface instance.

The Interface Name and the Series Number is used as a key to find the ISEC record.

Multiple ISEC records can be associated with one Interface instance.

Series key sets cannot be used if future keys are supported.



TRANSACTION SECURITY

Keys

Encryption Key Schemes

- BASE24-eps uses two types of encryption key schemes:
 - Symmetric key scheme
 - Asymmetric key scheme



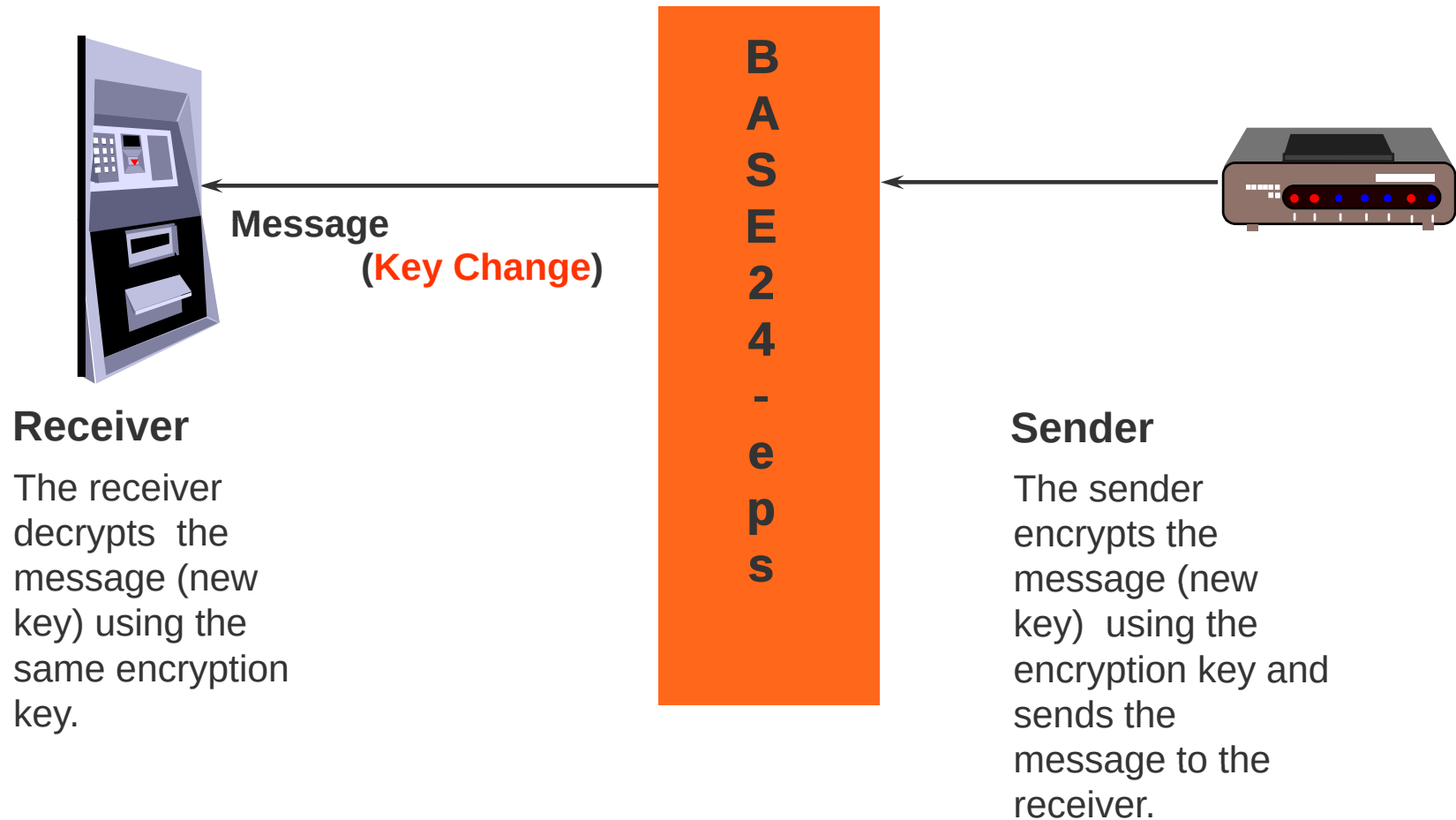
Symmetric Key Scheme

- Symmetric Key Scheme:
 - The same key used to encrypt is also used to decrypt.
 - Both entities share this same key
 - Cryptographic functions that support symmetric keys include:
 - Dynamic key change
 - New PIN/MAC encryption keys are exchanged programmatically. These new keys are encrypted for transmission using a shared KEK.
 - PIN encryption, Message Authentication, Card Verification, PIN translation



Symmetric Cryptography

Messages are encrypted by the sender and decrypted by the receiver using the same key.



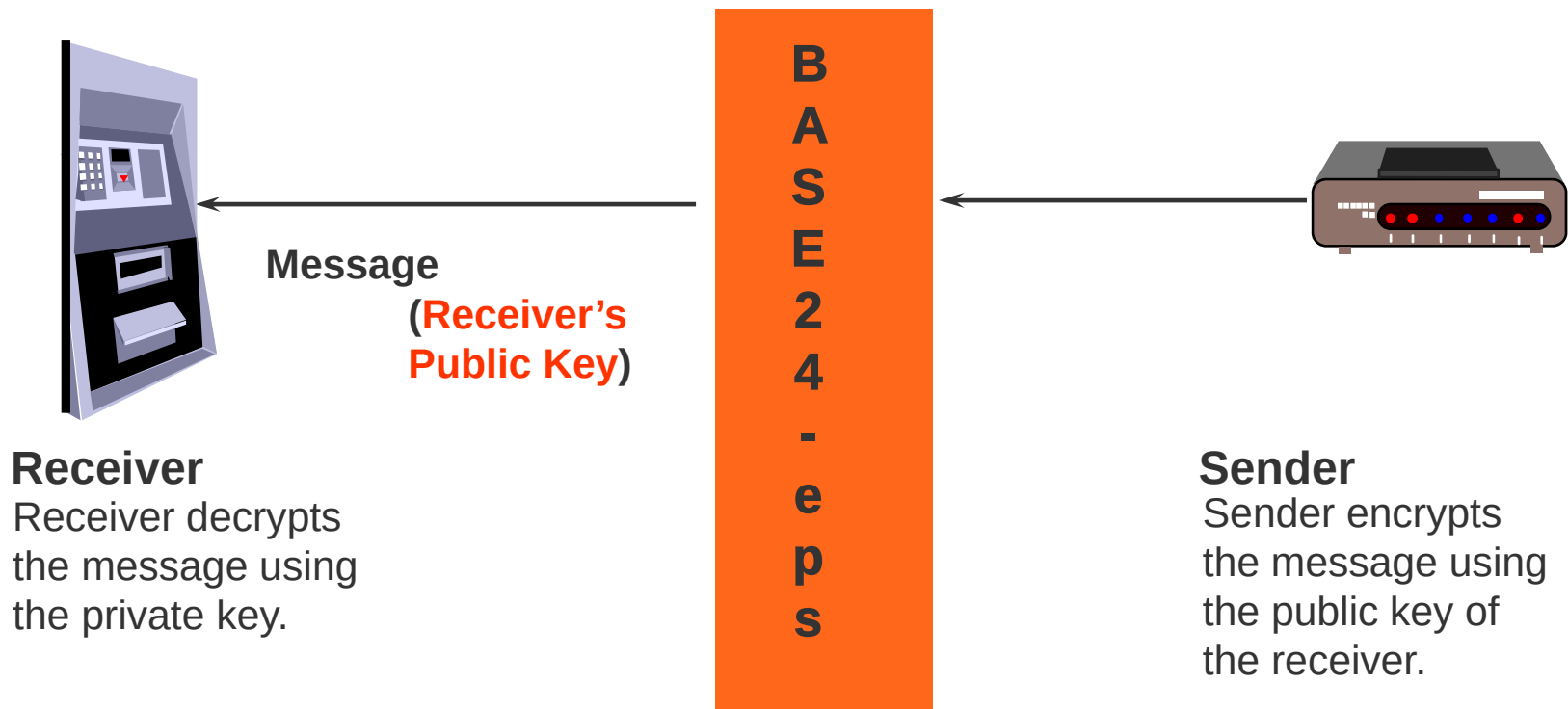
Asymmetric Key Scheme

- Public Key (asymmetric algorithms):
 - Public Key – The key used to encrypt data. This key can be widely distributed and does not need to be kept secret.
 - Private Key - A related, but different key used to decrypt data. This key must be kept secret.



Public Key Concept (Asymmetric)

Messages are encrypted by the sender using the receiver's public key and decrypted by the receiver using the receiver's private key.



Public Key Infrastructure (PKI)



MEETS THE CHALLENGE OF CHANGE

Confidential

ACI UNIVERSAL
PAYMENTS.



TRANSACTION SECURITY

Automated Key Distribution System (AKDS)

Automated Key Distribution System

- Used to download the master key to the ATM
- Based on the Rivest-Shamir-Adleman (RSA) scheme
- Improves key management practices
- Provides easier compliance with unique key per terminal mandates
- AKDS functions require the use of a hardware security module.
- Nouvelle Architecture Cryptographique (NAC) compliant version of AKDS available



Automated Key Distribution System

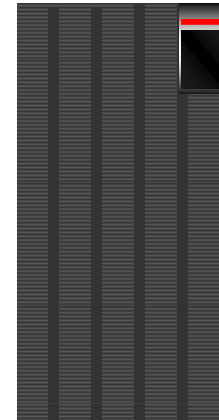
- AKDS functionality is supported by:
 - Diebold 912
 - NCR NDC
 - NCR Standard NDC
 - Shared 912
 - Shared NDC
 - Triton
 - Tidel



Automated Key Distribution System



The ATM's private key is stored within a tamper-resistant module.

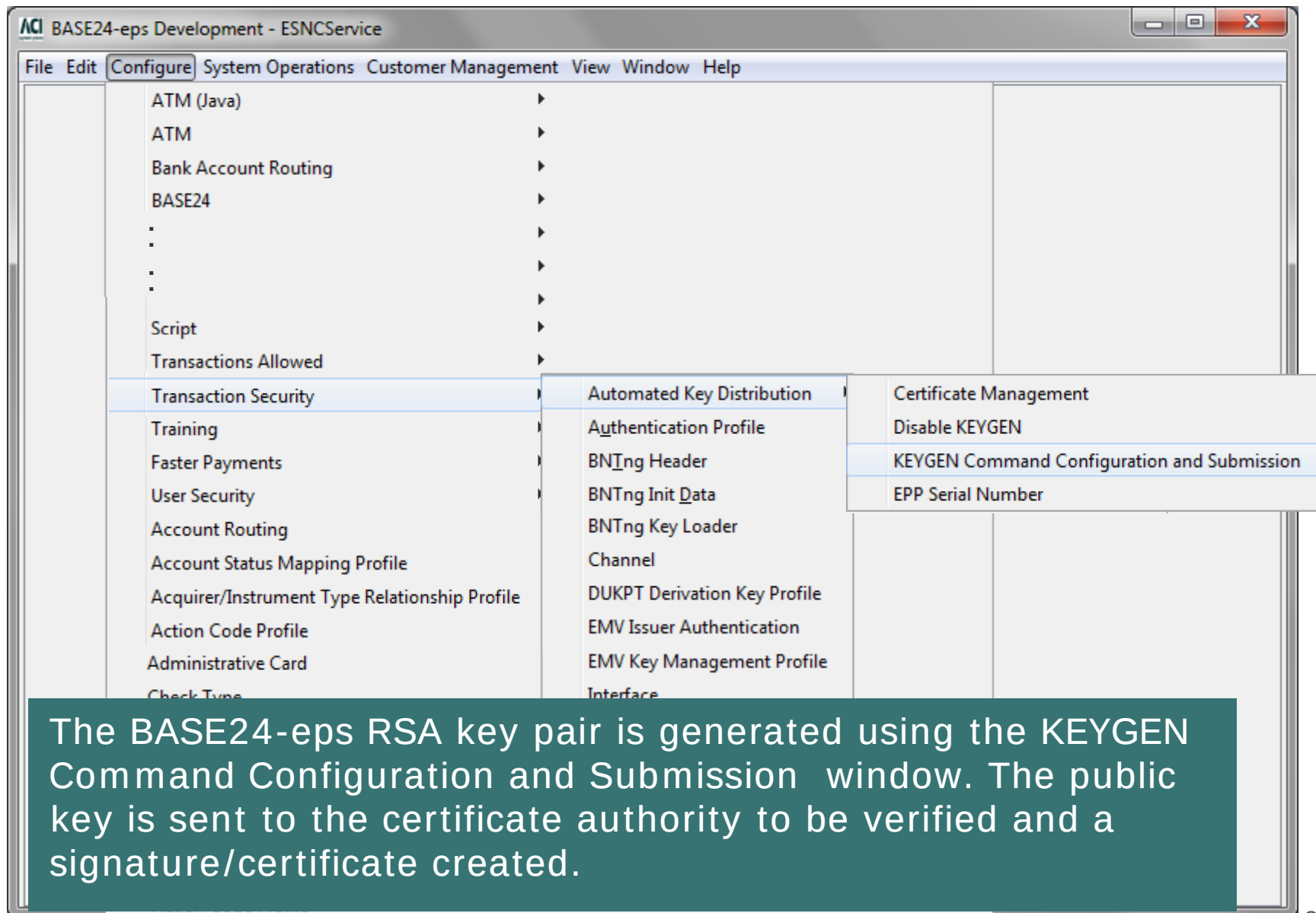


BASE24-eps' private key is stored in the database.

- The public keys are signed by a certificate authority's private key.
- The entities then authenticate each other and exchange their public keys.
- Once exchanged, these public keys are also stored.



BASE24-eps Key Pair - KEYGEN



BASE24-eps Key Pair - KEYGEN

ACI EPS Development - ESNCService

File Edit Configure System Operations Customer Management View Window Help

ACI KEYGEN Command Configuration and Submission

Command Configuration

Host Name: Device Type:

Common Name: Organization Name:

Override Current Key Indicator: ☐ Modulus Size:

Key Option: Effective Date:

Functions

Command Text

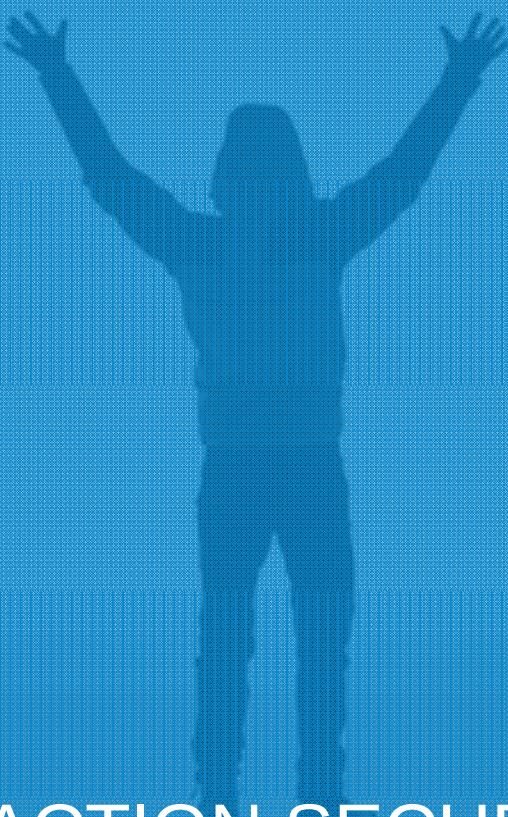
Response

Device Type

Select

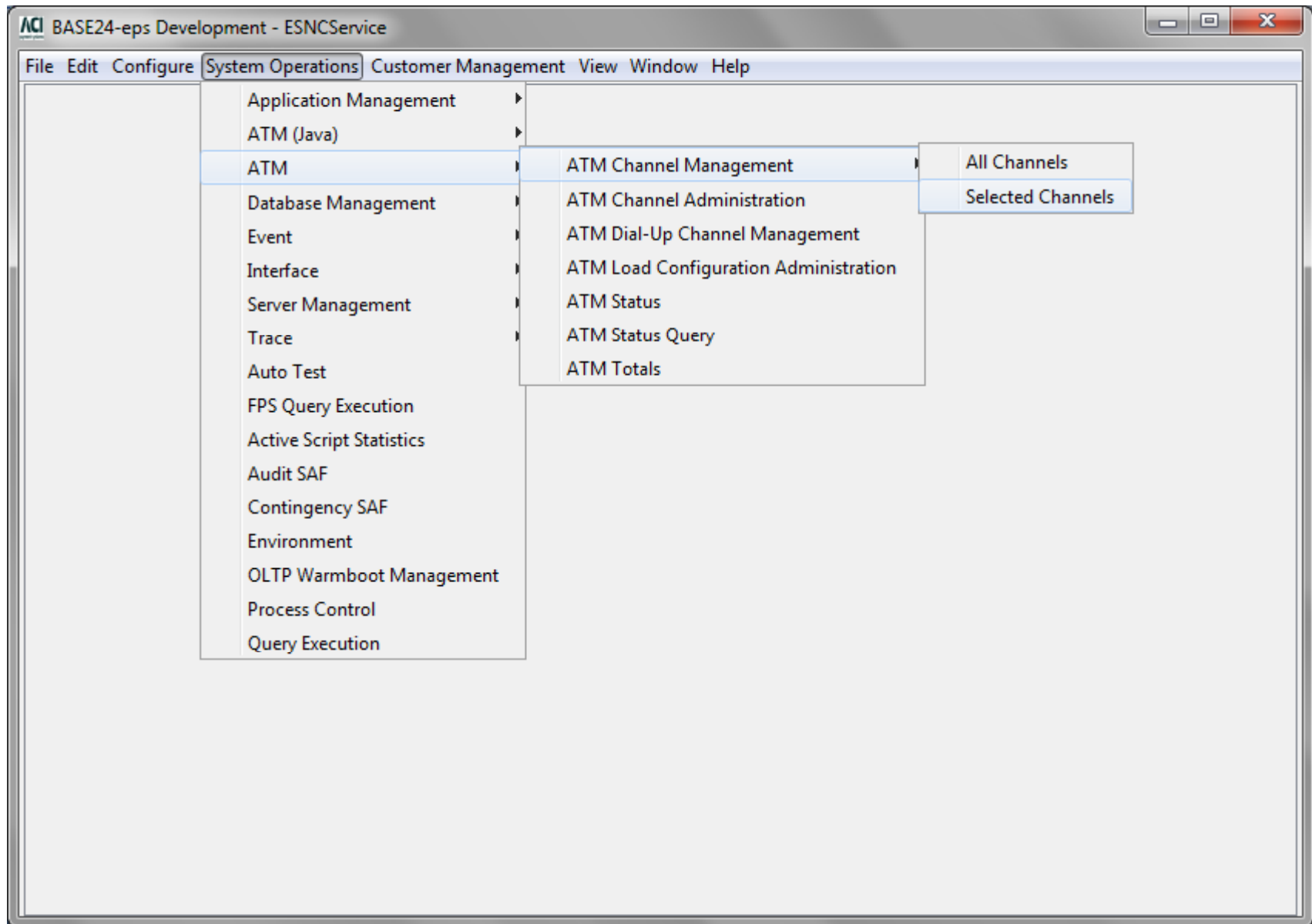
- NCR
- Diebold
- Wincor Signature Interface
- Wincor/NCR Signature
- Triton

287 - Desktop bounds(0,0,1101,790)



TRANSACTION SECURITY AKDS Configuration

Load Public Keys



Load Public Keys

ACI EPS Development - ESNCService

File Edit Configure System Operations Customer Management View Window Help

ACI ATM Channel Management

ATM Channel Search

Channel ID Channel Type

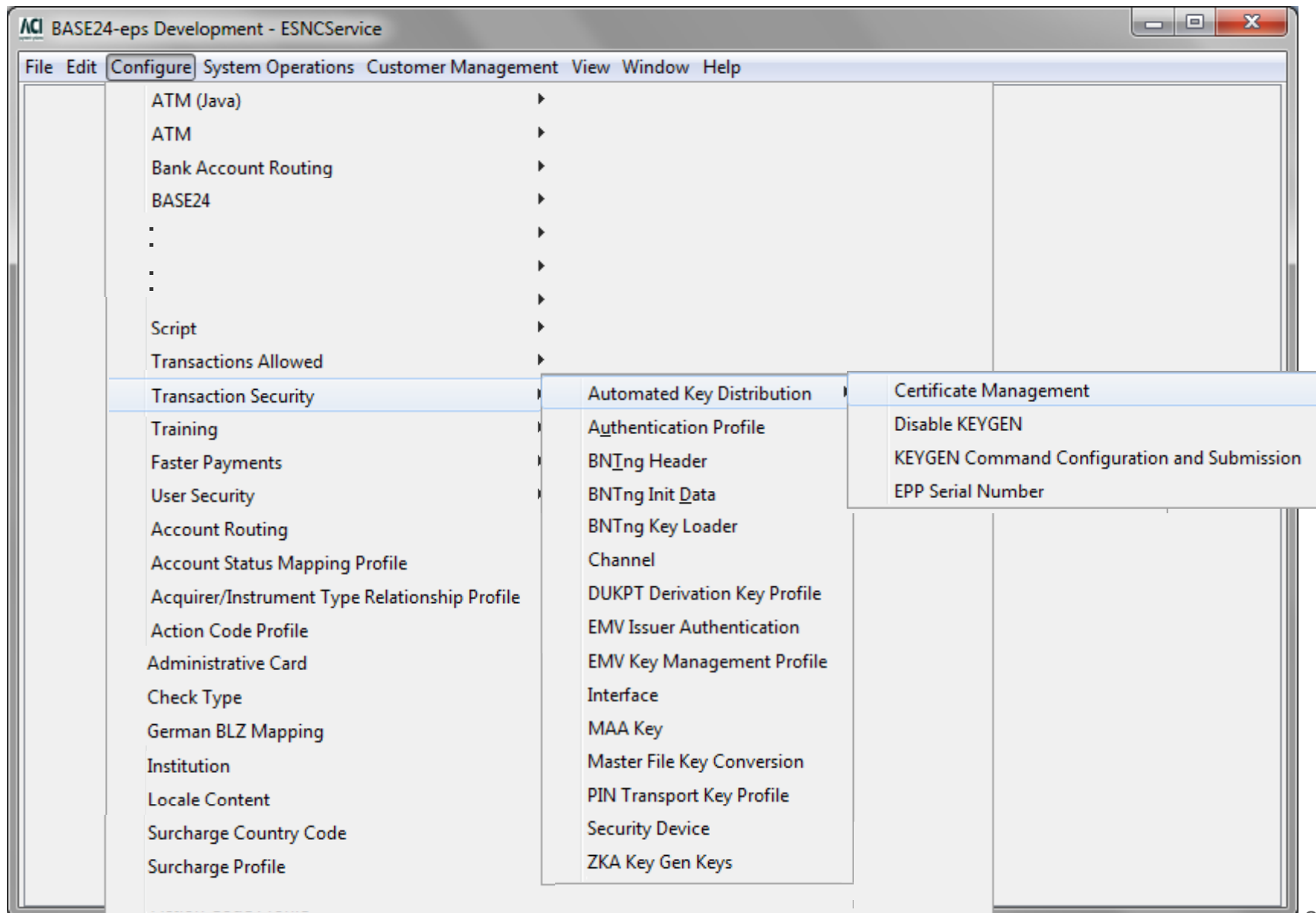
Institution ID Load Profile

Online Status Open Status

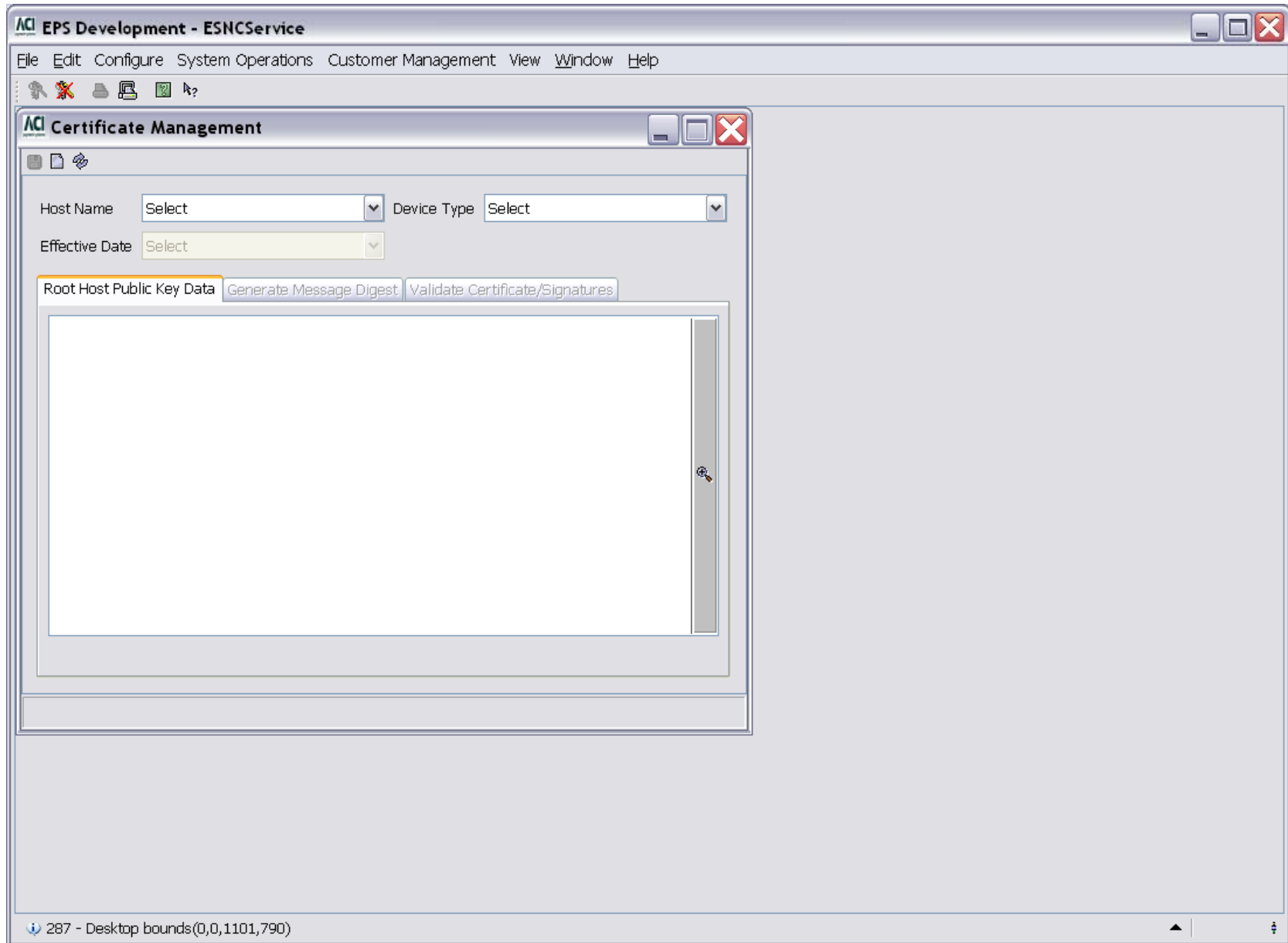
Channel ID	Institution ID	Channel Type	Load Profile	Online Status	Open Status	Command Queue Count	Action to Submit	Action Data	Load Group
------------	----------------	--------------	--------------	---------------	-------------	---------------------	------------------	-------------	------------

287 - Desktop bounds(0,0,1101,790)

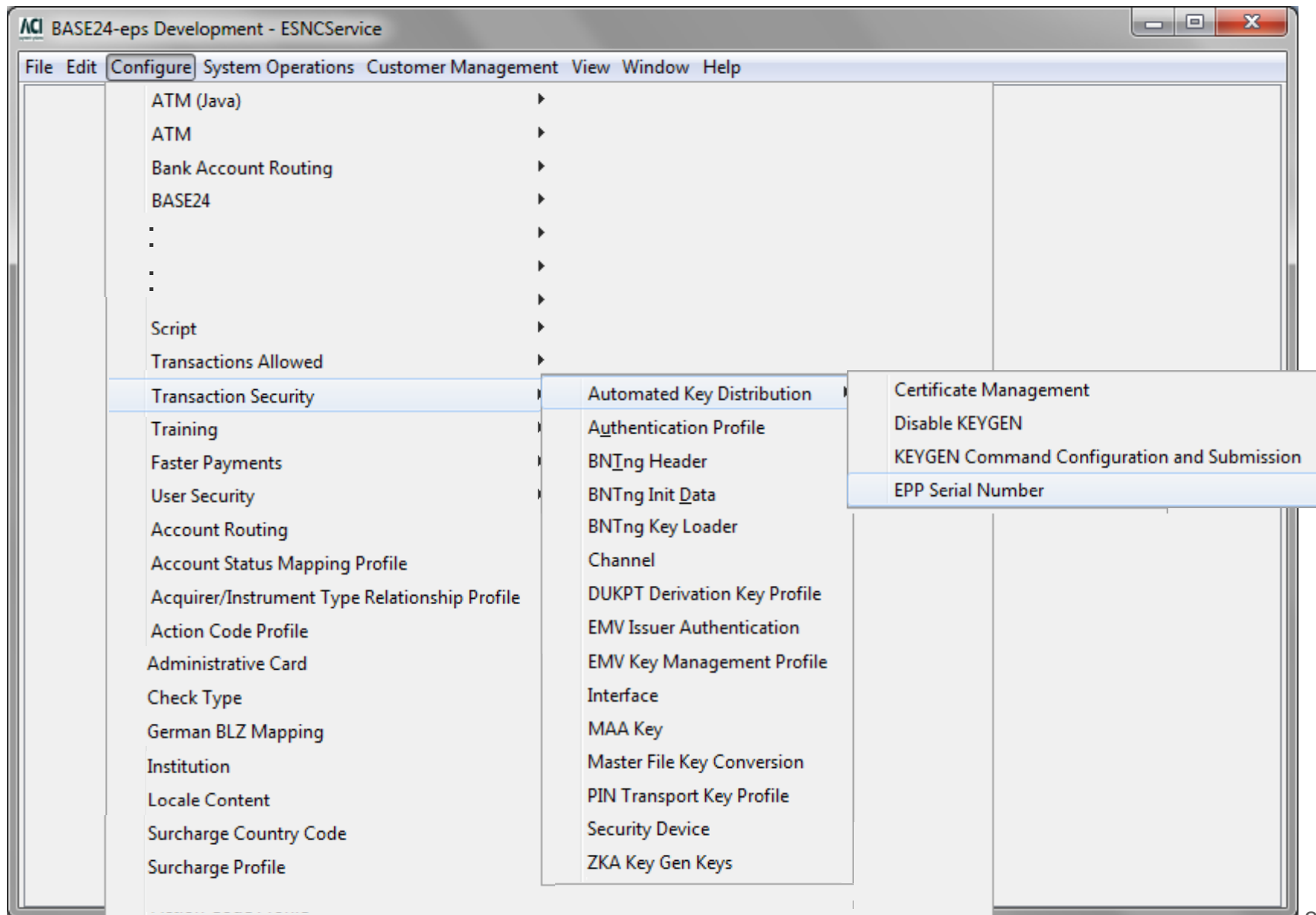
Certificate Management



Certificate Management



EPP Serial Number Assignment



EPP Serial Number Assignment

The screenshot shows a software window titled "EPS Development - ESNCServices" with a menu bar (File, Edit, Configure, System Operations, Customer Management, View, Window, Help) and a toolbar. Inside, a sub-window titled "EPP Serial Number Configuration" is active. It features a search section with fields for "Institution ID" (containing "Enter"), "Device Type" (a dropdown menu showing "Select"), "Serial Number From", and "Serial Number To". Below the search fields is a table with four columns: "Serial Number", "Channel ID", "Validation Date", and "Validation Time". The table has one empty row. The status bar at the bottom indicates "287 - Desktop bounds(0,0,1101,790)".

EPS Development - ESNCServices

File Edit Configure System Operations Customer Management View Window Help

EPP Serial Number Configuration

EPP Serial Number Search

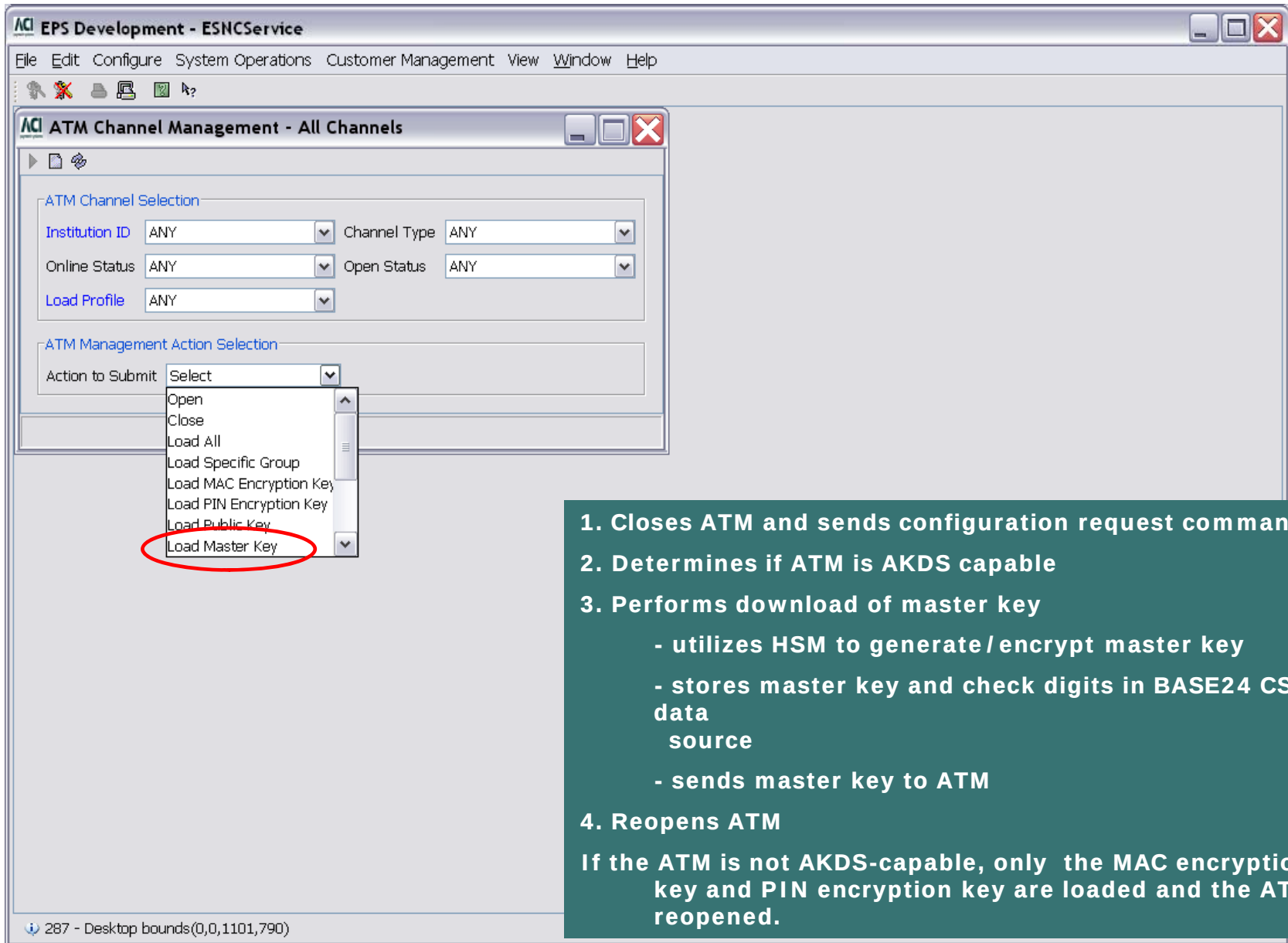
Institution ID Enter Device Type Select

Serial Number From Serial Number To

Serial Number	Channel ID	Validation Date	Validation Time

287 - Desktop bounds(0,0,1101,790)

Master Key Change



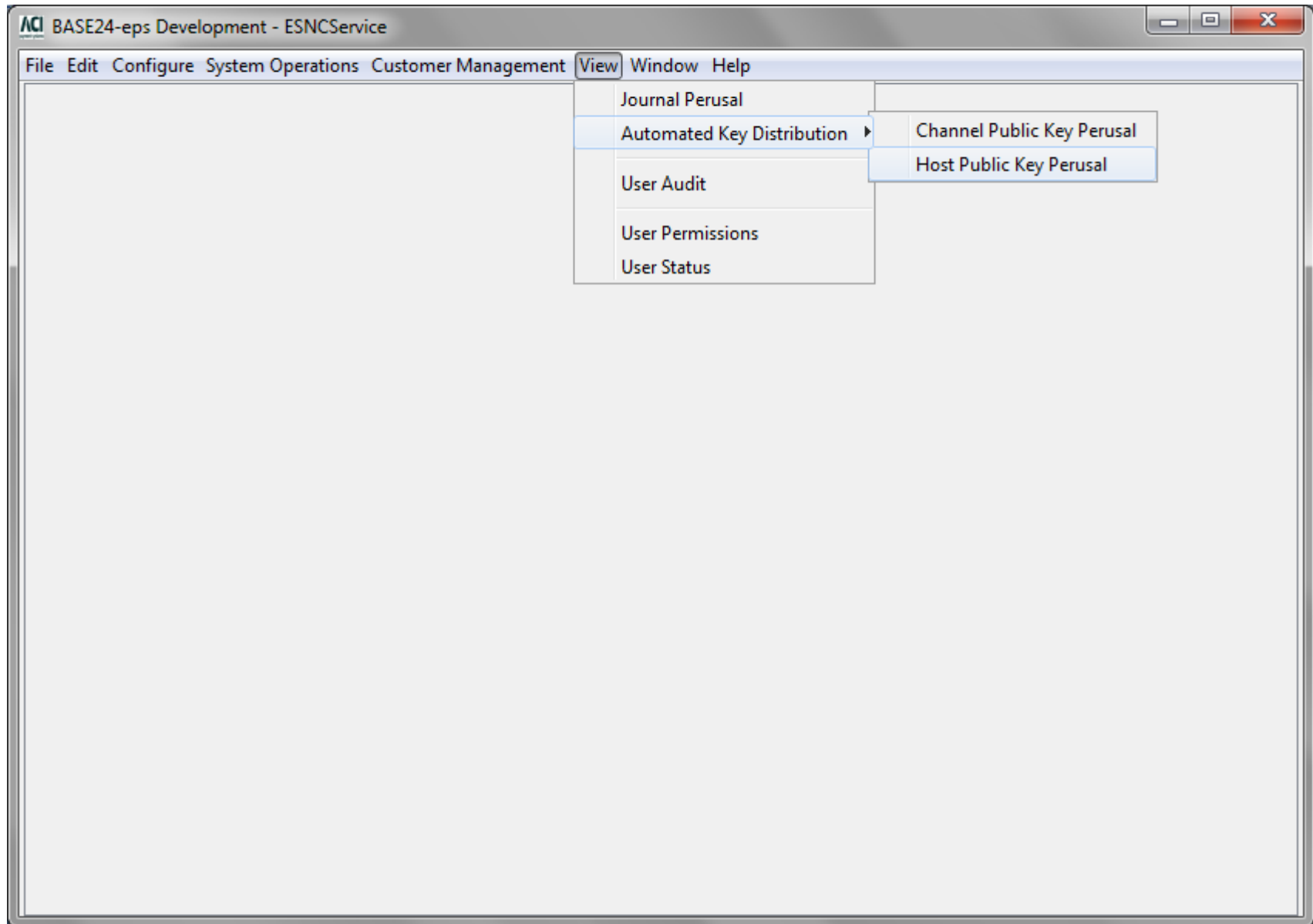
The screenshot shows the 'EPS Development - ESNCServices' application window. The 'ATM Channel Management - All Channels' sub-window is active. It contains two sections: 'ATM Channel Selection' with filters for Institution ID, Channel Type, Online Status, Open Status, and Load Profile, all set to 'ANY'; and 'ATM Management Action Selection' with a dropdown menu. The dropdown menu is open, showing options: Open, Close, Load All, Load Specific Group, Load MAC Encryption Key, Load PIN Encryption Key, Load Public Key, and 'Load Master Key', which is circled in red. A text box on the right lists the steps for the master key change process.

1. Closes ATM and sends configuration request command
2. Determines if ATM is AKDS capable
3. Performs download of master key
 - utilizes HSM to generate / encrypt master key
 - stores master key and check digits in BASE24 CSEC data source
 - sends master key to ATM
4. Reopens ATM

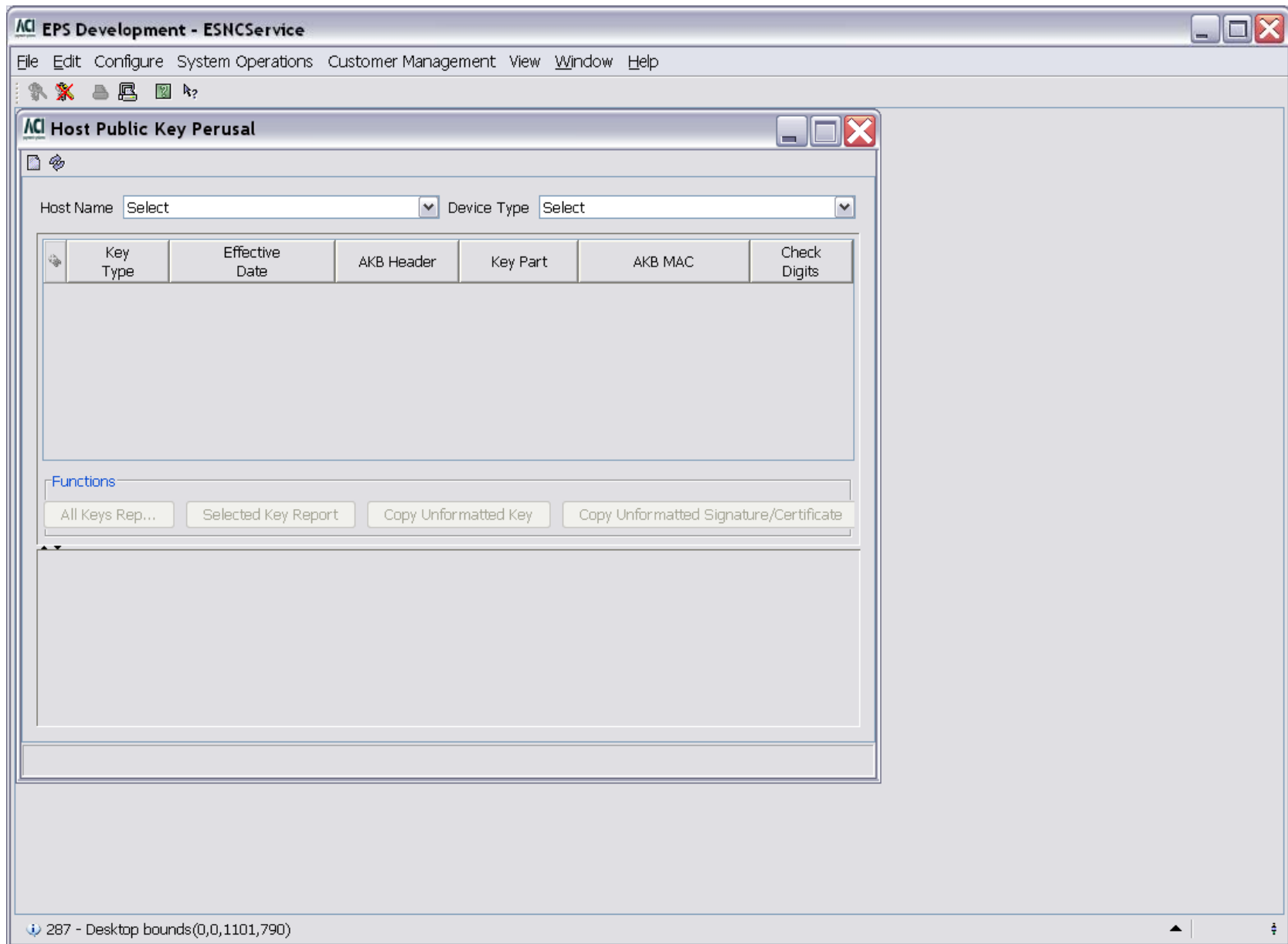
If the ATM is not AKDS-capable, only the MAC encryption key and PIN encryption key are loaded and the ATM is reopened.

287 - Desktop bounds(0,0,1101,790)

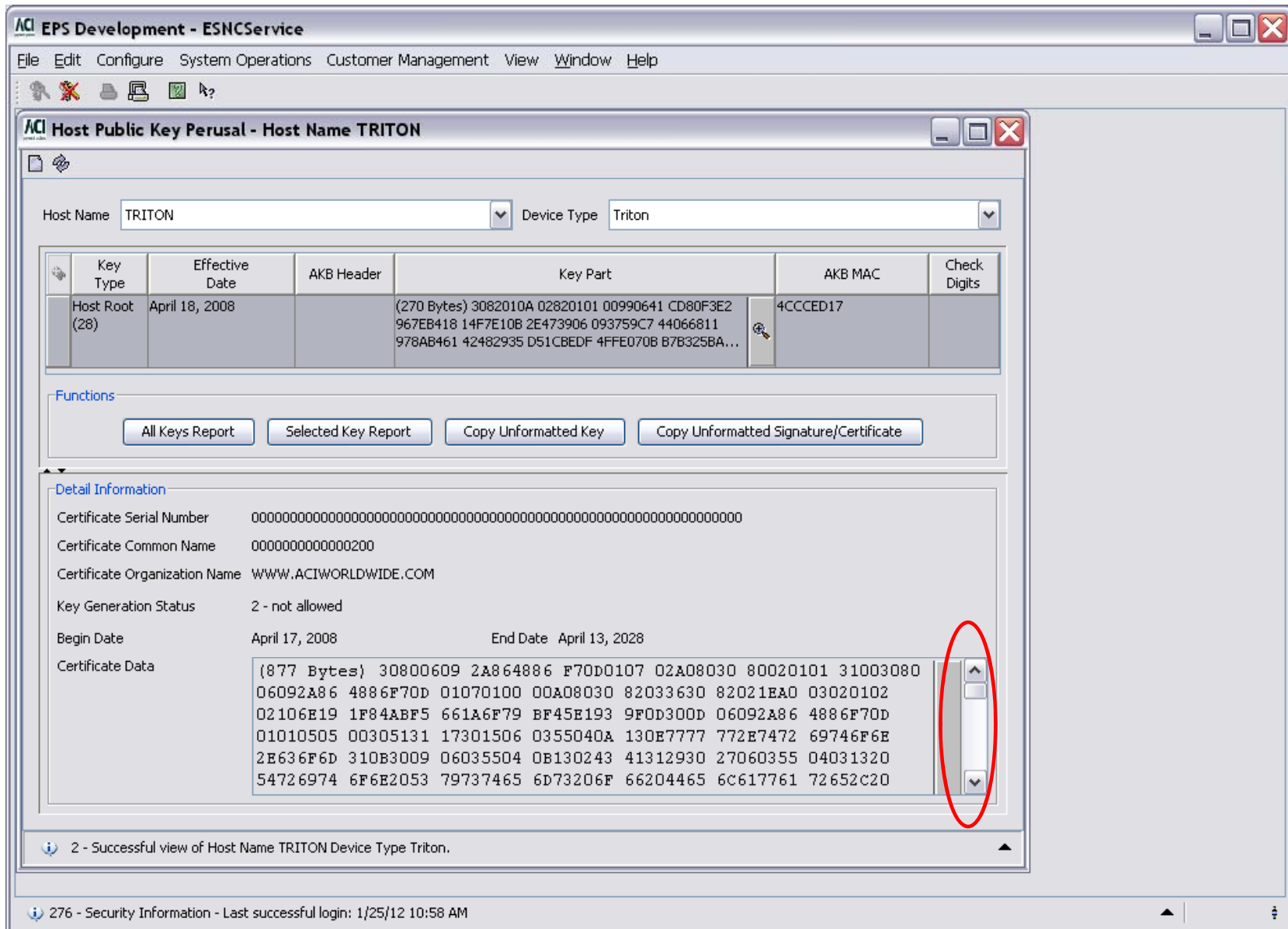
Host Public Key Perusal



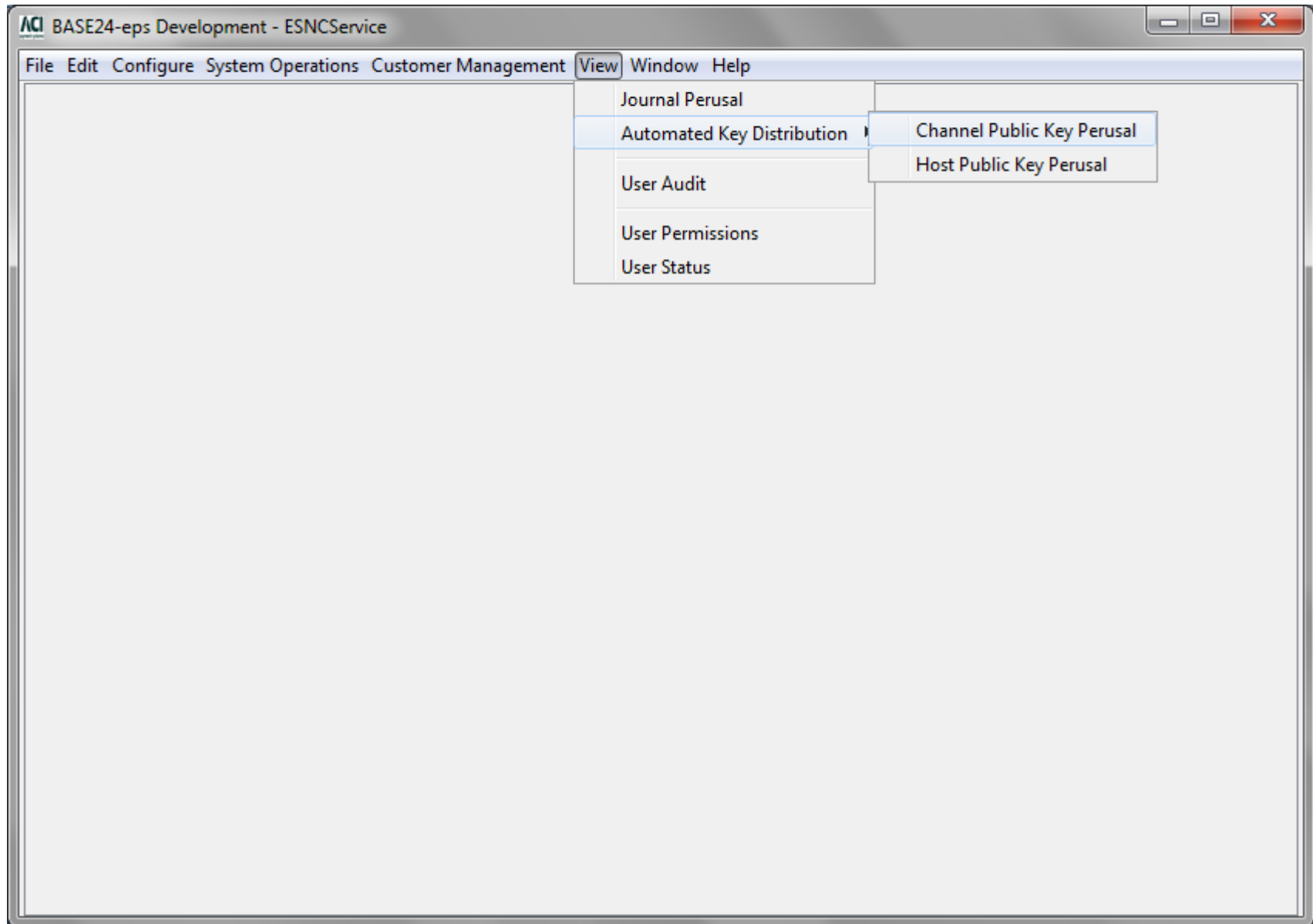
Host Public Key Perusal



Host Public Key Perusal



Channel Public Key Perusal



Channel Public Key Perusal

Channel ID: S74RTRITOMA Device Type: Triton

Key Type	AKB Header	Key Part	AKB MAC	Check Digits
EPP Signing Key		(270 Bytes) 3082010A 02820101 008370DD 192B361C F3BA4080 AC245922 E568A670 CE1A11DF 96389CB7 19A5701D D51F1534 8D1FDAFD 3F85D0A6 B457E819 52DBD9DF 32A163FB 62702942 DF71A154 5228F22B B132B9A3 5B872E34 3A8F556A 71238F43 BDE061D5 7B46AC16 7F46B038 D9B13C4A 7B489B9C DE31B907 8A778CF9 699A13F6 2F08CD71 FA6822FA 1080BB7C...	A47118CE.....	
EPP Encipherment Key		(270 Bytes) 3082010A 02820101 00F684E8 F9200E33 2B625803 C2EF0779 D42B581B 4ECFE1C2 F0BC946C 5D0F6A93 B61F67CE 370EFDA6 35F12328 82048DDC 5CDE54B1 638F53B1 81456180 5C8F53B1 638F53B1...	9ADAE6E5.....	

Functions:

View All Keys Report View Selected Key Report Copy Unformatted Key

Detail Information:

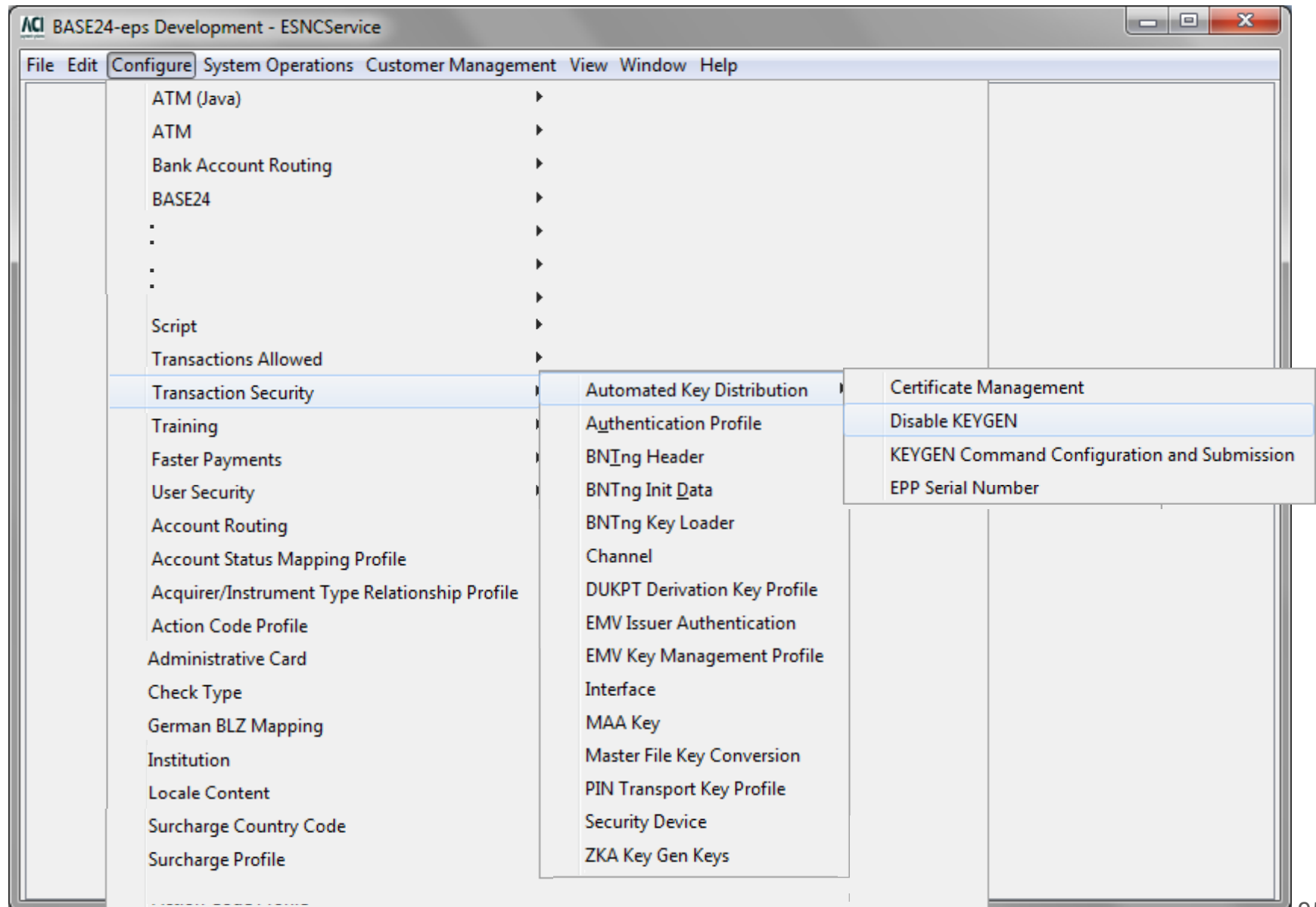
Begin Date: March 18, 2008 End Date: March 14, 2028

EPP Random Nonce: 0x0000000269044549 Host Random Nonce: 0x3233303831363036

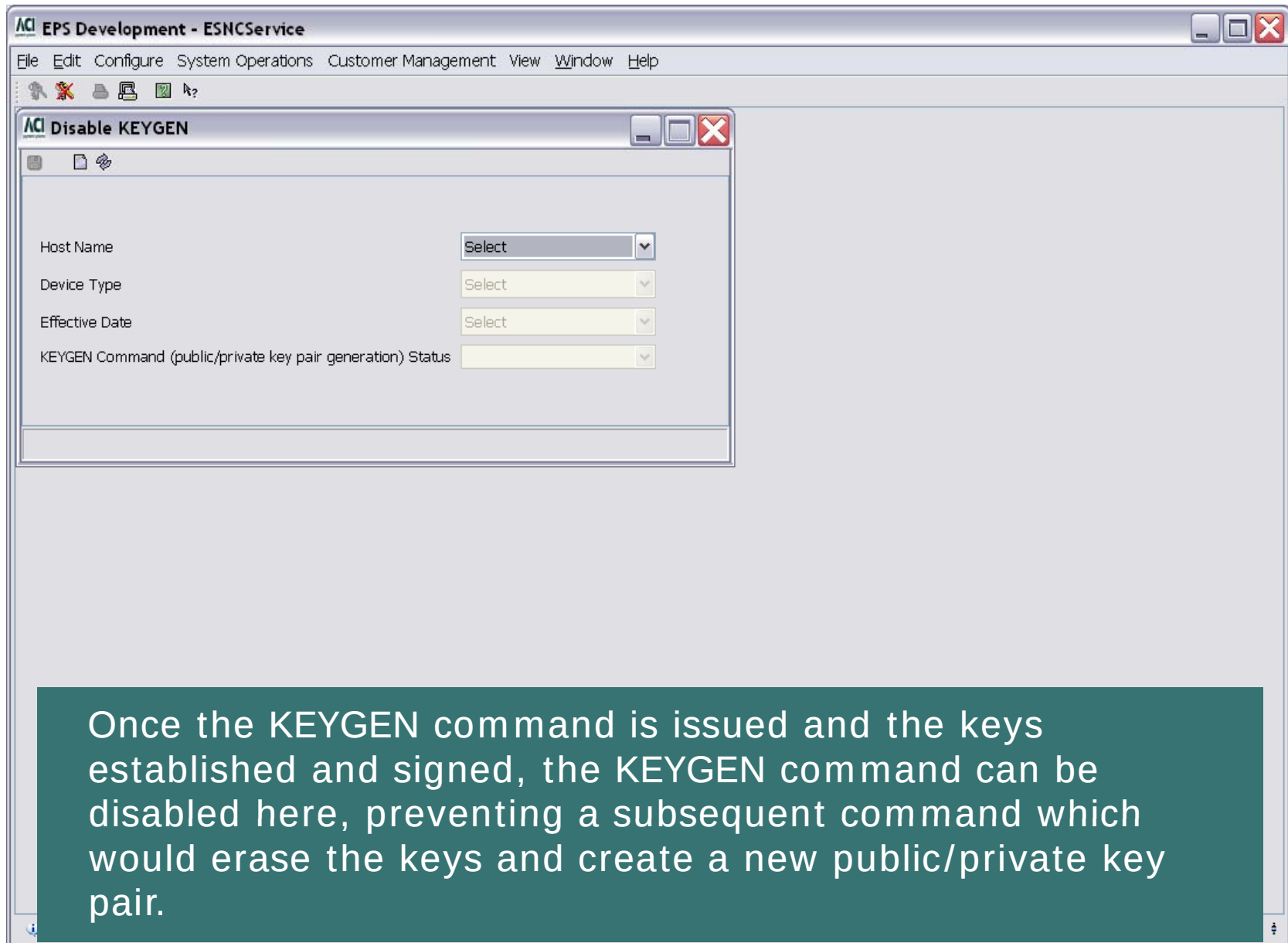
2 - Successful view of Channel ID S74RTRITOMA.

276 - Security Information - Last successful login: 1/25/12 10:58 AM

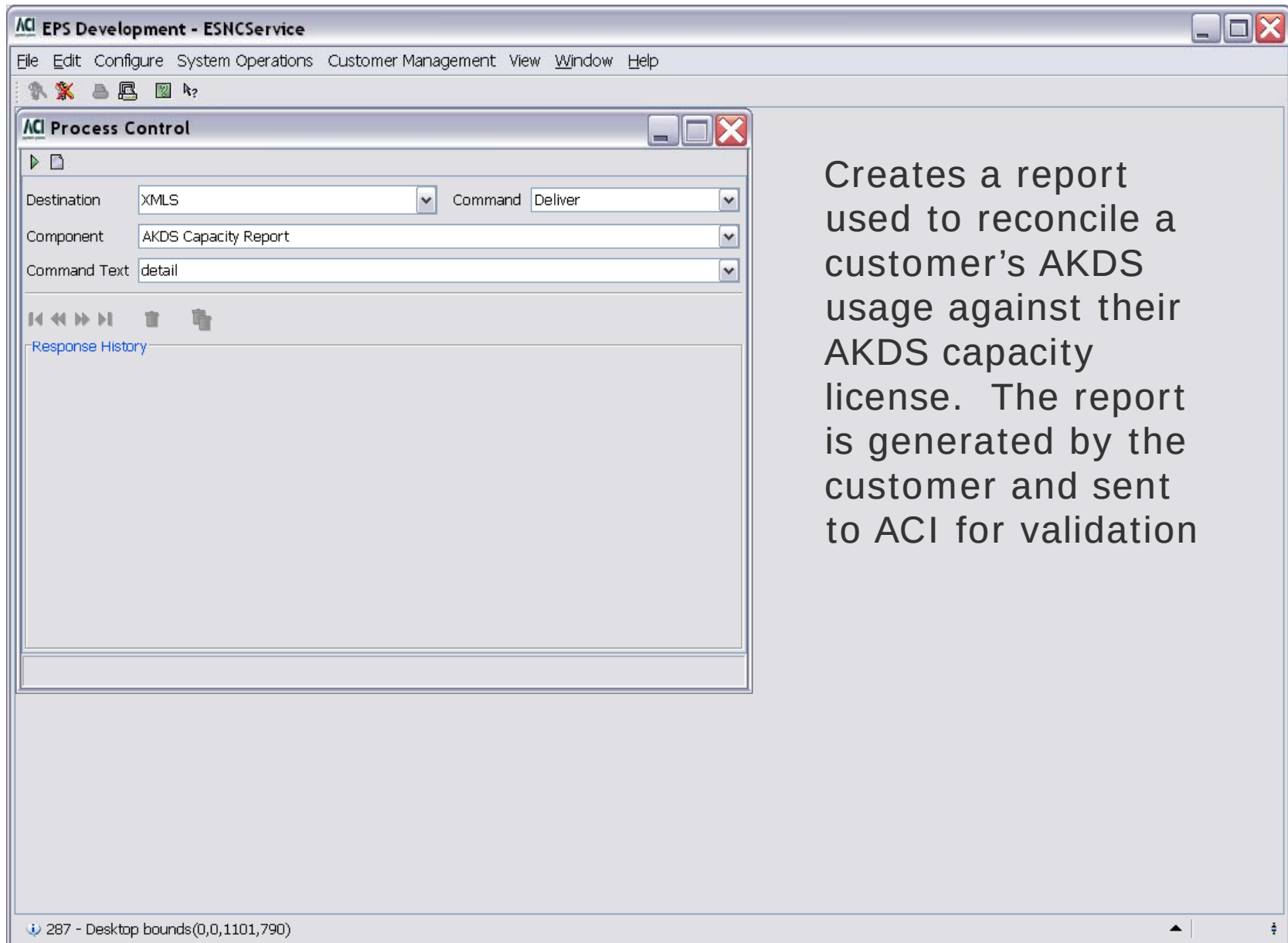
Disabling KEYGEN Functions



Disabling KEYGEN Functions



AKDS Capacity Report



Creates a report used to reconcile a customer's AKDS usage against their AKDS capacity license. The report is generated by the customer and sent to ACI for validation

AKDS Capacity Report

- The process reads sequentially through the Channel Public Key Security data source (detail option)

Customer Name - AKDS Capacity Report - Device Detail			
Date Run: 22 FEB 2007			
Device Ids:			
DIEBOLD001	NCR001	TERM01	TERM02
TERM03	TERM04	TERM05	TERM06
TERM07	TERM08	TERM09	TERM10
TERM100	TERM101	TERM102	TERM103
TERM104	TERM105	TERM106	TERM107
TERM108	TERM109	TERM11	TERM110
TERM111	TERM112	TERM113	TERM114
TERM115	TERM116	TERM117	TERM118
TERM119	TERM12	TERM120	TERM121
TERM122	TERM123	TERM124	TERM125
TERM126	TERM127	TERM128	TERM129
TERM13	TERM130	TERM131	TERM132
TERM133	TERM134	TERM135	TERM136
TERM137	TERM138	TERM139	TERM14
TERM140	TERM141	TERM142	TERM143
TERM144	TERM145		
Total Devices: 62			
Page 1			



ACI UNIVERSAL
PAYMENTS.



MEETS THE CHALLENGE OF CHANGE